Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions

with the Riemann Hypothesis as a corollary

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"... a 'viewpoint' by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally 'real'. It is when complementary viewpoints of a common reality are conjugated, that is, when our 'eyes' are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ... it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision."

— A. Grothendieck, *Récoltes et Semailles*

"The measure of our success is whether what we do enables people to understand and think more clearly and effectively about mathematics."

— W. Thurston, *On Proof and Progress in Mathematics*

Abstract

Let $\mathcal{R}$ be the commutative ring of slice-preserving slice-regular functions on the compactified octonions $\mathbb{O}^* = S^8$, under the regular $(*)$ product. Part 1 gathers the classical facts about the Riemann zeta function. Part 2 develops the slice-preserving theory, builds $\mathcal{R}$, places the octonionic zeta $\zeta_{\mathbb{O}^*}$ inside it, and proves the zero and equivalence theorems: the classically nontrivial zeros of ζ are residue-$\mathbb{C}$ zero 6-spheres, and the Riemann Hypothesis holds exactly when those spheres are concentric, sharing a single real centre.

Part 3 proves the main result, the *Concentricity Theorem*, for a section $A \in \mathcal{R}$ satisfying four conditions: (C1) meromorphic continuation with a single simple pole carried to $\infty = N$; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization whose spherical factors are the residue-$\mathbb{C}$ zeros; and (C4) infinitely many such zeros. The theorem says the residue-$\mathbb{C}$ zero 6-spheres of any such A are concentric.

The proof builds a concentricity detector from the section itself, out of groupoids, orbit–stabilizer, and categorical homotopy theory. Conditions (C1)–(C3) construct one distinguished exponential Möbius action on the projective base $\mathcal{B} = PGL(2,\mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$: the continued pole factor's logarithm λ_A is exponentiated in the diagonal matrix $\mathrm{diag}(e^{\lambda_A}, 1)$ and conjugated into the slice Möbius group, with the Euler product presenting the action on the half-space and the Weierstrass product presenting it at N. The slice-preserving normalization is G_2-equivariant on $\mathbb{O}^*$, and orbit–stabilizer positions this whole point-valued action at every object

of the base. Its sphere-world projection records the directions and Möbius transformations, while the A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ has the resulting normalized A-value states and their transports as its fibres. Its Grothendieck construction is $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$.

The C-residue system is the inverse-image action groupoid $\mathcal{R}_A$ selected inside $\mathcal{A}_A$, with fully faithful inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$, and hence a fully faithful inclusion of totals $\int \iota_A : \int_{\mathcal{B}} \mathcal{R}_A \hookrightarrow \mathcal{T}_A$. The comparison at the one north object, by the action's own morphisms and G_2, makes this inverse-image total transitive, therefore connected. Its component colimit is consequently a singleton. Every arrow of the residue total is an element of the one action with its logarithmic lift, together with an element of G_2; the real level is constant along every arrow, and the constant-functor theorem of the π_0 adjunction, instantiated at the certified representatives, reads one real value $c \in \mathbb{R}$ across the population. Thus every populated residue-$\mathbb{C}$ zero 6-sphere has centre c. The theorem asserts the shared centre; naming the number is the task of the corollaries.

Downstream corollaries translate residue-$\mathbb{C}$ zeros to classically nontrivial zeros and, for $\zeta_{\mathbb{O}^*}$, use its functional equation to identify $c = \frac{1}{2}$: the Riemann Hypothesis.

Contents

Introduction

The Riemann Hypothesis is usually met head-on, as a question about where the zeros of a single complex function fall along a line. This paper approaches it from the side. It asks instead whether a family of geometric objects, sitting a few dimensions up in the octonions, share a common centre, and shows that they must. The Riemann Hypothesis then follows, not as the thing we set out to prove, but as the shadow this larger fact casts back down onto the complex plane.

A nontrivial zero of the zeta function, seen from the complex line alone, is a lone point in a strip. Seen from the compactified octonions, that same zero opens into a six-dimensional sphere, and the critical-line question becomes a question of arrangement: are these spheres concentric, or scattered into different hyperplanes? It takes the whole continuum of imaginary directions in the octonions, read together, to see whether the alignment holds.

The paper is built to make that convergence visible and, at every step, checkable. Part 1 recalls the classical facts about the Riemann zeta function we will lean on. Part 2 develops the slice-preserving theory of functions on the octonions, builds the commutative ring $\mathcal{R}$ in which the octonionic zeta lives, and proves the reframing at the centre of everything: the classically nontrivial zeros are exactly the residue-$\mathbb{C}$ zero six-spheres, and the Riemann Hypothesis holds precisely when those spheres are concentric. Part 3 proves the main result, the Concentricity Theorem, with a small piece of machinery I think of as a concentricity detector. From the section itself we build one distinguished action on a projective base, position it on the whole continuum of Riemann spheres by orbit stabilizer, and read off, through categorical homotopy theory, whether the zero-spheres land in a single connected piece. They do; and for the octonionic zeta, that is the Riemann Hypothesis.

No one can picture an eight-dimensional graph in their head, but tools from categorical homotopy theory, like the total-object Grothendieck construction and standard colimit arguments, let us probe this structure and reason about it. The argument is also formalized in Lean, so that each step can be checked; the reader can not only verify the steps but, I hope, come to understand for themselves why the Concentricity Theorem is true and why the Riemann Hypothesis follows.

Part I
Classical Background

1 The Riemann zeta function

The Riemann zeta function is classical, and we take its basic theory as given. We recall only what the construction uses: the meromorphic continuation and functional equation, the Euler product over the primes, the Hadamard product over the zeros, and Hardy's theorem. We then read ζ in compactified form, as a map to the Riemann sphere, which is the form in which it will extend to the octonions.

Theorem 1.1 (Meromorphic continuation and functional equation; Riemann [13])**.** *The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$: holomorphic as a $\mathbb{C}$-valued function on $\mathbb{C} \setminus \{1\}$, with a single simple pole at $s = 1$. Equivalently, read into the Riemann sphere, the same function is a holomorphic map on the whole complex plane,*

$$\zeta : \mathbb{C} \longrightarrow \mathbb{C}^*, \qquad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \qquad \zeta(1) = \infty,$$

the pole becoming the regular value ∞. Holomorphy is on the domain $\mathbb{C}$ in both readings: with infinitely many zeros, ζ is not rational, so the domain point ∞ carries an essential singularity. (The *later extension of the* domain *to* $\mathbb{C}^*$ *assigns a marked value at* ∞*, Definition 1.6 with Remark 1.7: an assignment of geometry and value, made where the classical statement ends.) The completed function* $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ *is entire and satisfies*

$$\xi(s) = \xi(1-s), \qquad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ *is a nontrivial zero of* ζ *then so are* $\bar{\rho}$*,* $1-\rho$*, and* $1-\bar{\rho}$*. The trivial zeros of* ζ *are at* $s = -2, -4, \ldots$*; the nontrivial zeros lie in the strip* $0 < \operatorname{Re} s < 1$*. The Riemann Hypothesis asserts that every nontrivial zero has* $\operatorname{Re} s = \frac{1}{2}$*.*

Theorem 1.2 (Infinite Euler product; [15, Ch. 1])**.** *For* $\operatorname{Re} s > 1$*,*

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - p^{-s}\right)^{-1} = \exp\Bigl(\sum_{p}\sum_{k\geq 1} \tfrac{1}{k}\, p^{-ks}\Bigr),$$

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\operatorname{Re} s > 1$*.*

Theorem 1.3 (Infinite Hadamard product; [15, Ch. 2])**.** ξ *is entire of order* 1 *with infinitely many zeros, which are exactly the nontrivial zeros* $\{\rho\}$ *of* ζ*, and admits the Hadamard factorization*

$$\xi(s) = \xi(0)\prod_{\rho}\Bigl(1 - \tfrac{s}{\rho}\Bigr)e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Corollary 1.4 (Infinitude of the nontrivial zeros)**.** ζ *has infinitely many nontrivial zeros.*

Proof. Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of ζ. □

Theorem 1.5 (Hardy [11])**.** *Infinitely many nontrivial zeros of* ζ *lie on the line* $\operatorname{Re} s = \frac{1}{2}$*.*

The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction, ζ is taken in compactified form, with values on the Riemann sphere $\mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2$ (the one-point compactification of $\mathbb{C}$ [12]). The classical facts above are retained as stated.

Definition 1.6 (Compactified classical zeta)**.** $\zeta_{\mathbb{C}^*} : \mathbb{C}^* \to \mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$ for $s \in \mathbb{C} \setminus \{1\}$; $\zeta_{\mathbb{C}^*}(1) = \infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty) := 1$, an assigned marked value (Remark 1.7), selected by the one genuine limit $\zeta(s) \to 1$ as $\operatorname{Re}(s) \to +\infty$ (the p-test: only the $n = 1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C} \setminus \{1\}$, its sole pole being $s = 1$; at the domain point ∞ no analytic claim is made. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s}) = \overline{\zeta_{\mathbb{C}^*}(s)}$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C} \setminus \{1\}$ where $\zeta_{\mathbb{C}^*} = \zeta$.

Remark 1.7 (The value at infinity is a marked point, not an extension)*.* No analytic claim is made, or could be made, at the domain point ∞. A function holomorphic on the whole Riemann sphere is rational, and ζ is not: it has an essential singularity at ∞, tending to 1 as $\operatorname{Re} s \to +\infty$ while passing through the trivial zeros along the negative real axis, so no assigned value makes $\zeta_{\mathbb{C}^*}$ so much as continuous there. Definition 1.6 therefore performs two distinct acts, and only the first is analytic.

Regarding the meromorphic ζ as a holomorphic map into the target sphere, with $\zeta_{\mathbb{C}^*}(1) = \infty$, is the classical sphere-valued reading of a meromorphic function. Assigning $\zeta_{\mathbb{C}^*}(\infty) := 1$ supplies the value the compactified setting consumes as data: the point $\infty = N$ is shared by every slice Riemann sphere (Definition 2.4), it is where the pole's value lands, and it is an object of the projective base the transport machinery of Part 3 runs on (Definition 9.23), so compactified value paths are read at it. The assignment is not free. Slice preservation forces the value to lie in every slice sphere at once, and the intersection of the slice spheres is $\mathbb{R} \cup \{N\}$; the limit along the one region through which the construction approaches ∞ — the right half-plane, where $\zeta \to 1$ genuinely — then selects 1. The essential singularity is visible only along directions the construction never travels. What is consumed downstream is the value, never smoothness: no statement in this paper uses continuity or differentiability of $\zeta_{\mathbb{C}^*}$ at ∞. The same discipline governs the whole class of Part 3: an A-section has infinitely many zeros, hence is not rational, hence has an essential singularity at N; accordingly every analytic hypothesis of Definition 10.4 is stated at finite points, and the value at N is carried as bare data (in the Lean development, the field `valueAtInfinity`). This is the marked-extension device of Remark 3.6: the compactification supplies geometry — the pole's value becomes a point of the target sphere, and all slice spheres share the single point N — never an extension. A companion distinction completes the picture. At N, Part 3 positions a group action carrying analytic content: the slice-preserving content the hypotheses supply over the octonions. There N serves as an object of the projective base, as the point of each value sphere at which the distinguished Möbius element arrives in Weierstrass form, and as the shared point of the slice spheres. The analysis that produces the element read there happens at the finite pole — continuation through the simple pole — and its output is positioned at N by the orbit representative. No step of the construction evaluates the section analytically at the domain point ∞: the position and the marked point are one geometric point playing two roles, as a group element's position and as a function's domain point, and only the first is ever active in the construction.

Lemma 1.8 (Compactification preserves the zero set)**.** *$Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.*

Proof. On $\mathbb{C} \setminus \{1\}$ the two functions agree, so they share all zeros there. The two adjoined values lie outside the zero set: $\zeta_{\mathbb{C}^*}(1) = \infty$ is a pole and $\zeta_{\mathbb{C}^*}(\infty) = 1$. Hence compactification preserves the zero set exactly. □

Part II

Slice-Preserving Theory, the Ring $\mathcal{R}$, and the Equivalence

2 Octonions and slices

The octonions are an 8-dimensional real division algebra, the largest of the normed division algebras. This section fixes what we need from them: the algebra itself; Artin's theorem, that any two octonions generate an associative subalgebra, so that powers and slicewise complex analysis make sense; and the complex slices $\mathbb{C}_v$, one for each imaginary direction $v \in S^6$, together with their compactifications, the slice Riemann spheres. All the slices share the real axis and a single point at infinity, the north pole N; that shared geometry is what the later construction turns around.

The octonions $\mathbb{O}$ have basis $\{1, e_1, \ldots, e_7\}$ with each $e_i^2 = -1$.

Definition 2.1. The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

An octonion $s \in \mathbb{O}$ decomposes as $s = \sigma + w$ with $\sigma = \mathrm{Re}(s) \in \mathbb{R}$ and $w = \mathrm{Im}(s) \in \mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$; the unit imaginary octonions form $S^6 \subset \mathrm{Im}(\mathbb{O})$.

Theorem 2.2 (Artin [14])**.** *In an alternative algebra, the subalgebra generated by any two elements is associative.*

Corollary 2.3. *Powers s^n are well-defined for any $s \in \mathbb{O}$.*

Definition 2.4 (Complex slices and slice Riemann spheres)**.** For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$-algebra isomorphism $\varphi_v : \mathbb{C} \to \mathbb{C}_v$, $i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9], holomorphicity on each complex slice separately, as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin's theorem each slice $\mathbb{C}_v$ is an associative subalgebra where classical complex analysis applies unchanged).

Definition 3.1 (Slice regularity; [5, Def. 5.1.1])**.** Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \to \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

Theorem 3.2 (Representation Formula; [5, Thm. 5.1.7])**.** *Let f be slice regular on an axially symmetric slice domain Ω. Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2} v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.*

Theorem 3.3 (Identity Theorem; [5, Cor. 5.1.9])**.** *Let f be slice regular on an axially symmetric slice domain Ω. If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω.*

Theorem 3.4 (Extension Theorem; [5, Thm. 5.1.5])**.** *Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \to \mathbb{C}$ holomorphic. There is a unique slice regular $\mathrm{ext}(g) : \Omega_U \to \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U,\ v \in S^6\}$ with $\mathrm{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0.*

Definition 3.5 (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16])**.** A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

(i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);

(ii) its representation-formula components are $\mathbb{R}$-valued: writing $f(\sigma+\gamma v)=\alpha(\sigma,\gamma)+v\,\beta(\sigma,\gamma)$ (Theorem 3.2), one has $\alpha,\beta:\mathbb{R}^2\to\mathbb{R}$, equivalently the inducing stem function $F=F_1+\iota F_2$ has $\mathbb{R}$-valued components F_1,F_2;

(iii) f preserves two distinct slices; equivalently $f(\Omega\cap\mathbb{R})\subseteq\mathbb{R}$ when Ω is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti– Stoppato [7, Rem. 1.8]).

Remark 3.6 (Compactified extension). Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains $\Omega\subseteq\mathbb{O}$ (and $U\subseteq\mathbb{C}$). We use them on the compactified $\mathbb{O}^*=S^8$ and on the slice Riemann spheres $\mathbb{C}_v^*=S_v^2$ (GPV's slice Riemann manifolds [17]). The passage $\mathbb{O}\rightsquigarrow\mathbb{O}^*$, $\mathbb{C}_v\rightsquigarrow\mathbb{C}_v^*$ adjoins only the shared point $\infty=N$; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

Slice-preserving theory already gives the tools to build the octonionic zeta function. On each slice $\mathbb{C}_v$ we read the compactified classical ζ through the isomorphism $\varphi_v:\mathbb{C}^*\to\mathbb{C}_v^*$, and the representation formula glues these slice copies into a single function on all of $\mathbb{O}^*$. The one thing to check is that the two identifications of a slice with $\mathbb{C}$ agree, so that $\zeta_{\mathbb{O}^*}$ is well-defined; they do, because conjugating the input and reversing the identification cancel. The pole of ζ at $s=1$ is carried to the north pole N. At the shared point N itself the function and the geometry are defined but nothing more is claimed: the value there is the assigned marked value 1 of Remark 1.7 — as real $s\to+\infty$, $\zeta(s)\to1$, the one genuine limit — and neither continuity nor differentiability at N is asserted or used by the argument.

Definition 4.1 (Octonionic zeta)**.** Here N denotes the *north pole* of S^8: the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^*=\mathbb{O}\cup\{\infty\}\cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty=N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^*=S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*}:\mathbb{O}^*\to\mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v:\mathbb{C}^*\xrightarrow{\sim}\mathbb{C}_v^*=S_v^2$ (the $\mathbb{R}$-algebra isomorphism $\mathbb{C}\to\mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty)=N$; GPV [17], Prop. 4.1/Thm. 4.2):

(i) for $s\in\mathbb{O}\setminus\mathbb{R}$, with $w=\mathrm{Im}(s)$, $v=w/|w|\in S^6$, $\gamma=|w|>0$, $\sigma=\mathrm{Re}(s)$, so $s=\sigma+\gamma v$: $\zeta_{\mathbb{O}^*}(s):=\varphi_v(\zeta_{\mathbb{C}^*}(\sigma+i\gamma))$;

(ii) for $s\in\mathbb{R}\setminus\{1\}$: $\zeta_{\mathbb{O}^*}(s):=\zeta_{\mathbb{C}^*}(s)=\zeta(s)\in\mathbb{R}$;

(iii) for $s=1$ (the simple pole, Definition 1.6): $\zeta_{\mathbb{O}^*}(1):=\infty=N$;

(iv) for $s=\infty=N$: $\zeta_{\mathbb{O}^*}(\infty):=\zeta_{\mathbb{C}^*}(\infty)=1$, the assigned marked value of Remark 1.7: as real $s\to+\infty$, $\zeta(s)\to1$, and the definition asserts the value only — neither continuity nor differentiability of $\zeta_{\mathbb{O}^*}$ at N is claimed here or used anywhere in the argument.

Proposition 4.2 (Well-definedness)**.** $\zeta_{\mathbb{O}^*}:\mathbb{O}^*\to\mathbb{O}^*$ *is well-defined.*

Proof. For $s \in \mathbb{O} \setminus \mathbb{R}$ the decomposition $s = \sigma + \gamma v$ with $\gamma > 0$, $v \in S^6$ is forced ($\gamma = |\operatorname{Im} s|$, $v = \operatorname{Im} s/|\operatorname{Im} s|$), so the only ambiguity is the identification of the slice $\mathbb{C}_v = \mathbb{C}_{-v}$ with $\mathbb{C}^*$. That slice carries two such identifications, φ_v ($i \mapsto v$) and φ_{-v} ($i \mapsto -v$), under which s reads as $\sigma + i\gamma$ resp. $\sigma - i\gamma$; we must check $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = \varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma - i\gamma))$. By conjugate symmetry $\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$ ([15, Ch. 2]; Definition 1.6), writing $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$ gives $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$: the two sign reversals, one from conjugating the input, one from the reversed identification, cancel. The remaining cases carry no ambiguity: for real $s \neq 1$, $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ for every v; at $s = 1$ the value is the single point $\infty = N$; and at $s = \infty = N$ the value is $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$, again the same for every v since each φ_v fixes $\mathbb{R}$ pointwise. Well-definedness at N is a statement about the value, not about regularity: the assigned marked value 1 lies in $\mathbb{R}$, which every slice sphere contains, so it is unambiguous across all slices, and the function and the geometry are thereby defined at N while continuity and differentiability there are neither claimed nor used (Remark 1.7). □

5 $\mathcal{R}$ is a commutative ring

The slice-preserving sections form a commutative ring $\mathcal{R}$, with $\zeta_{\mathbb{O}^*}$ among its elements. Its multiplication is the regular $(*)$ product, which descends from the Cayley–Dickson construction of $\mathbb{O}$ and from slice regularity; on slice-preserving functions it restricts on each slice to the ordinary pointwise product (Wang), and is therefore commutative. The ring axioms, and the integral-domain property, follow slicewise.

We work on $\mathbb{O}^*$; write $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$ for the slice Riemann sphere (Definition 2.4).

Definition 5.1 (The ring of slice-preserving functions under the regular product)**.** $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-preserving functions $f : \mathbb{O}^* \to \mathbb{O}^*$ (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, slice regular away from their marked points — the poles, where the value is ∞, and the shared point N (Remark 1.7) — equipped with pointwise addition and the regular $(*)$ product.

Commutativity of $\mathcal{R}$ rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

Theorem 5.2 (Regular product on slice-preserving functions; [16, Rem. 2.11])**.** *For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).*

Proposition 5.3. *$(\mathcal{R}, +, *)$ is a commutative ring with identity $1 \in \mathcal{R}$.*

Proof. Closure under $+$ is pointwise. For $*$: fixing a slice $\mathbb{C}_K$, the product reduces to pointwise multiplication on $\mathbb{C}_K$ by Theorem 5.2, which is commutative because $\mathbb{C}_K$ is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity. □

Proposition 5.4 ($\mathcal{R}$ is an integral domain)**.** *If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.*

Proof. $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$ in $\mathbb{C}_K \cong \mathbb{C}$ (Theorem 5.2). The restrictions $f|_{\Omega_K}$, $g|_{\Omega_K}$ are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes. □

6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and I-independent lift, and the winding lift. Each is established in the literature on the uncompactified $\mathbb{O}$ and used here on $\mathbb{O}^* = S^8$ and the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

Theorem 6.1 (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23])**.** *The exponential* $\exp : \mathbb{O} \to \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k \geq 0} q^k/k!$ *(powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \qquad x, y \in \mathbb{R},\ I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ *depends only on the real part, and* $\exp$ *is nowhere zero.*

Theorem 6.2 (The logarithm manifold $E_{\mathbb{O}}^+$; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3])**.** *Let* $T : \mathbb{O} \to \mathbb{O} \times \operatorname{Im} \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y,\ Iy)$, *and* $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$. *The* logarithm manifold *is* $E_{\mathbb{O}}^+ := T(\mathbb{O}^+)$, *the semi-helicoidal hypercomplex* 8*-manifold; equivalently it is the image of the exponential graph, parametrised by the* $E_{\mathbb{O}}^+$-exponential

$$E : \mathbb{O} \longrightarrow E_{\mathbb{O}}^+ \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \qquad E(x + Iy) = (\exp(x + Iy),\ Iy) = (e^x \cos y + Ie^x \sin y,\ Iy),$$

an immersion and a diffeomorphism, with inverse the $E_{\mathbb{O}}^+$-logarithm $L : E_{\mathbb{O}}^+ \to \mathbb{O}$, $L(q, p) = \log|q| + p$ *(here* $p = \operatorname{Arg}(q)$*). Writing* π *for the projection to the first factor,* $\pi \circ E = \exp$ *(Theorem 6.1). The manifold* $E_{\mathbb{O}}^+$ *is an adapted blow-up of* $\mathbb{O}$ *along the real axis. Its projection* $\pi : E_{\mathbb{O}}^+ \to \mathbb{O} \setminus \{0\}$ *has the degenerate real fibres responsible for the failure of covering and openness. The blow-up unwraps the multivalued logarithm into the single-valued* L*, and a branch of the hypercomplex logarithm is* $\log_{\mathbb{O}} = L \circ (\pi|_{\Omega})^{-1}$ *on any path-connected* $\Omega \subset E_{\mathbb{O}}^+$ *on which* $\pi|_{\Omega}$ *is injective.*

Lemma 6.3 (The degenerate set of the exponential; its fibre over a real value)**.** *With* π *the first-factor projection,* $\pi \circ E = \exp$ ([17, Rem. 5.2(a)], the commuting triangle)*. Its spherical* degenerate set *makes the projection an adapted blow-up with failures of covering and openness* ([17, Rem. 5.2(b)])*. The direction* $I(q)$ *becomes singular at every point of* $\mathbb{R}$ ([21, Rem. 2.1])*.*

Proved from the slice form. *For* $r > 0$*,*

$$\exp^{-1}(-r) = \left\{ \log r + I(2k{+}1)\pi :\ I \in S^6,\ k \in \mathbb{Z} \right\} :$$

by Theorem 6.1, $\exp(x+Iy) = e^x \cos y + Ie^x \sin y = -r$ *forces* $e^x \sin y = 0$*, hence* $y \in \pi\mathbb{Z}$*; negativity of the value forces* $y = (2k{+}1)\pi$ *and* $x = \log r$*; the direction* I *is unconstrained. The fibre is thus indexed by the single real* level $\log r = \log|-r|$ *and, within it, by the odd winding indices* $2k{+}1$ *carried as band data (the winding lift, Theorem 6.5; [21, Cor. 5.21]). The fibre formula also appears as Preface motivation in [17] (p. 972); the slice-form derivation above proves it.*

Definition 6.4 (Slice restriction and the I-independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2])**.** A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \qquad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \ \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ of Definition 2.4. The *I-independent lift* is the fact that this holomorphic stem is the *same* for every v: there is one holomorphic $A_\circ : \mathbb{C}^* \to \mathbb{C}^*$ with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_\circ \circ \varphi_v^{-1} \qquad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \to \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$-valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad F_1, F_2 \ \mathbb{R}\text{-valued}.$$

It is this single I-independent slice stem that makes a slice-preserving section G_2-equivariant, hence blind to the G_2-orbit direction (Definition 9.2).

Theorem 6.5 (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11])**.** *Let $\gamma : [a, b] \to \mathbb{O}\backslash\{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation).* A continuation of the hypercomplex logarithm along γ *is a path $\tilde\gamma : [a, b] \to \mathbb{O}$ with $\exp \circ \tilde\gamma = \gamma$, and a* lift of γ *to the logarithm manifold $E_{\mathbb{O}}^+$ (Theorem 6.2) is a path $\Gamma : [a, b] \to E_{\mathbb{O}}^+$ with $\mathrm{pr}_1 \circ \Gamma = \gamma$:*

$$\begin{tikzcd} & \mathbb{O} \arrow[d, leftarrow, "\exp"] \\ {[a,b]} \arrow[ur, "\tilde\gamma"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd} \qquad \begin{tikzcd} & E_{\mathbb{O}}^+ \arrow[d, "\mathrm{pr}_1"] \\ {[a,b]} \arrow[ur, "\Gamma"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

The two data are equivalent*: a lift Γ exists if and only if a continuation $\tilde\gamma$ exists, and they correspond by*

$$\tilde\gamma = L \circ \Gamma, \qquad \Gamma = (\exp \circ \tilde\gamma,\ \operatorname{Im} \tilde\gamma),$$

where L is the $E_{\mathbb{O}}^+$-logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop *$\gamma : S^1 \to \mathbb{O} \setminus \{0\}$ the lift is taken on the universal cover $\exp : i\mathbb{R} \to S^1$, as a continuous $\Gamma : i\mathbb{R} \to E_{\mathbb{O}}^+$ with $\mathrm{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):*

$$\begin{tikzcd} i\mathbb{R} \arrow[r, "\Gamma"] \arrow[d, "\exp"'] & E_{\mathbb{O}}^+ \arrow[d, "\mathrm{pr}_1"] \\ S^1 \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

Proposition 6.6 (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22])**.** *The lift of Theorem 6.5 is governed by the companion structure of γ.*

(i) Uniqueness (tameness). *γ is* tame *when it admits a* unique *companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).*

(ii) Existence and closure. *A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and* if it exists, the lift is itself a loop *([21, Cor. 5.13]; pinned against the arXiv text; the earlier citation to Cor. 5.22 is superseded).*

(iii) Winding number. *For an untwisted tame loop with even circular signature σ^c, the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).*

7 Slice preservation and symmetries

With the ring in hand, we check that $\zeta_{\mathbb{O}^*}$ actually lives in it, and record the symmetry that will organize its zeros. Slice preservation says $\zeta_{\mathbb{O}^*}$ keeps each slice inside itself; together with slice regularity this makes it a section of $\mathcal{R}$. The symmetry is the action of the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$, which fixes the real axis and rotates the imaginary directions S^6, and $\zeta_{\mathbb{O}^*}$ is equivariant under it. This forces every non-real zero to fill a whole 6-sphere, the geometry the next section reads off.

Theorem 7.1 (Slice preservation)**.** *$\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.*

Proof. Fix $v \in S^6$ and $s \in \mathbb{C}_v^*$. If $s \in \mathbb{C}_v \setminus \mathbb{R}$ then $s = \sigma \pm \gamma v$ and $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$ (Definition 4.1(i)). If $s \in \mathbb{R}\setminus\{1\}$ then $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$; while $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$ and $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$. In every case the value remains in $\mathbb{C}_v^*$. □

Theorem 7.2 ($\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$)**.** $\zeta_{\mathbb{O}^*} \in \mathcal{R}$.

Proof. On each slice the restriction is the compactified classical zeta read through the isomorphism φ_v, namely $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$, holomorphic into $\mathbb{C}_v^*$ away from the single pole. Equivalently $\zeta_{\mathbb{O}^*} = \mathrm{ext}(\zeta)$ is the slice-regular extension of the holomorphic, conjugation-symmetric ζ on $\mathbb{C} \setminus \{1\}$ (Theorem 3.4, [5, Thm. 5.1.5]); hence $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (slice regularity is slicewise holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving, $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$. A slice-regular, slice-preserving function $\mathbb{O}^* \to \mathbb{O}^*$ is by definition a section of $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$ (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Its zeros (Theorem 8.2) make $\zeta_{\mathbb{O}^*}$ a nonunit. □

Definition 7.3 (The automorphism group G_2)**.** $G_2 = \mathrm{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$-algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\mathrm{Im}\,\mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g\cdot(\sigma+\gamma v) = \sigma + \gamma\, g(v)$.

Theorem 7.4 (G_2-equivariance)**.** *$\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.*

Proof. For $s \in \mathbb{R}$, $g \cdot s = s$ and $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$ is fixed, so both sides equal $\zeta_{\mathbb{O}^*}(s)$. For a non-real $s = \sigma + \gamma v$ ($v \in S^6$, $\gamma > 0$), g carries the slice $\mathbb{C}_v$ to the slice $\mathbb{C}_{g(v)}$ by the isometric relabelling $\varphi_{g(v)} = g \circ \varphi_v$ (Definition 7.3). Since $\zeta_{\mathbb{O}^*}$ is defined slicewise by the same $\zeta_{\mathbb{C}^*}$ read through φ_v (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}\big(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)\big) = g\big(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))\big) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

□

Theorem 7.5 ([3])**.** *G_2 acts transitively on S^6 with stabilizer* SU(3)*:* $\mathrm{SU}(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.

Remark 7.6 (The G_2-action extends to $\mathbb{O}^* = S^8$). The action above is stated on $\mathbb{O}$, but $\mathcal{H}_1$ (Part 3) runs on the compactified $\mathbb{O}^* = S^8$. Each $g \in G_2$ fixes the real axis pointwise (an automorphism fixes 1, hence all of $\mathbb{R}$) and acts on $\mathrm{Im}\,\mathbb{O} = \mathbb{R}^7$ as an element of $SO(7)$; in particular g is a linear isometry of $\mathbb{R}^8 = \mathbb{O}$, hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ fixing the point at infinity $\infty = N$ (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus $G_2 \curvearrowright \mathbb{O}^*$ fixes the compactified real axis $\mathbb{R} \cup \{\infty\}$, a great circle through N, pointwise, and acts on each direction-sphere S^6 exactly as in Theorem 7.5. This is the action on which $\mathcal{H}_1$ is built, with $\mathcal{R}$ its ring of invariants over $\mathbb{O}^*$.

8 Zero geometry and the equivalence

This section is the bridge the whole paper is built to cross. It shows that the zeros of $\zeta_{\mathbb{O}^*}$ are exactly the classical nontrivial zeros, now spread out: each nontrivial zero ρ of ζ becomes a whole 6-sphere of zeros of $\zeta_{\mathbb{O}^*}$, its residue-$\mathbb{C}$ zero-sphere, sitting at real part $\operatorname{Re}\rho$. The Riemann Hypothesis, that every ρ has real part $\frac{1}{2}$, becomes the statement that all these 6-spheres share the same real part, that they are concentric. From here on we work with the spheres.

Theorem 8.1 (Zero Equivalence Theorem). *Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,*

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \quad \Longleftrightarrow \quad \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.8 these are exactly the nontrivial zeros of the classical ζ.

Proof. By Definition 4.1(i), $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$. The slice identification $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^* = S_v^2$ is an isomorphism of slice Riemann spheres (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended by $\varphi_v(\infty) = N$; GPV [17]). An isomorphism is injective and sends 0 to 0, so $\varphi_v(z) = 0 \iff z = 0$; with $z = \zeta_{\mathbb{C}^*}(\rho)$ this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in $\mathbb{C}_v^*$ so that φ_v^{-1} applies on the non-real side.) The final clause is Lemma 1.8. $\square$

Theorem 8.2 (Nontrivial zeros are infinitely many disjoint 6-spheres). *For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set*

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma\, S^6 \subset \mathbb{O}.$$

Then:

(i) S_ρ is *a 6-sphere, the round sphere of radius γ centred at the real point σ, equivalently the G_2-orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser* SU(3)*, Theorem 7.5, Baez [3]).*

(ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar\rho}$;

(iii) the nontrivial-zero set is the disjoint union $Z^{\mathrm{nt}}_{\mathbb{O}^*} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar\rho\}$, with distinct pairs giving disjoint spheres;

(iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).

Proof. (i) The map $v \mapsto \sigma + \gamma v$ is the affine map $x \mapsto \sigma + \gamma x$ of $\operatorname{Im}\mathbb{O} \cong \mathbb{R}^7$ restricted to S^6; with $\gamma > 0$ its image is the round 6-sphere of radius γ centred at σ. By G_2-equivariance (Theorem 7.4) and transitivity of G_2 on S^6 (Theorem 7.5), $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma\, g(v_0) : g \in G_2\} = S_\rho$. So S_ρ *is* S^6, scaled by γ, translated to σ, realized as a single G_2-orbit; no embedding or diffeomorphism is invoked.

(ii) For any $v \in S^6$, $\zeta_{\mathbb{O}^*}(\sigma+\gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$ by the Zero Equivalence Theorem 8.1. And $S_{\bar\rho} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$, since $v \mapsto -v$ is a bijection of S^6.

(iii) For "$\subseteq$": a non-real zero $s = \sigma + \gamma v$ of $\zeta_{\mathbb{O}^*}$ forces $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$ by Theorem 8.1, so $s \in S_\rho$ with $\rho = \sigma + i\gamma$ nontrivial (Lemma 1.8); "$\supseteq$" is (ii). Disjointness: if $\sigma + \gamma v = \sigma' + \gamma' w$, uniqueness of the real/imaginary parts gives $\sigma = \sigma'$ and $\gamma v = \gamma' w$, whence $\gamma = \gamma'$ (taking norms, $|v| = |w| = 1$) and $\{\rho', \bar\rho'\} = \{\rho, \bar\rho\}$; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). $\square$

Theorem 8.3 (The equivalence: concentricity $\Leftrightarrow$ RH)**.** *The following are equivalent:*

(a) the Riemann Hypothesis;

(b) the nontrivial-zero 6*-spheres* $\{S_\rho\}$ *of* $\zeta_{\mathbb{O}^*}$ *(Theorem 8.2) are* concentric*: they share a single common centre, necessarily a point of the real axis.*

When they hold, the common centre is $\frac{1}{2}$*.*

Proof. Each S_ρ is centred at the real point σ_ρ, where $\rho = \sigma_\rho + i\gamma_\rho$ (Theorem 8.2(i)).

$(a) \Rightarrow (b)$: under RH every $\sigma_\rho = \frac{1}{2}$, so all spheres share the centre $\frac{1}{2}$.

$(b) \Rightarrow (a)$: suppose the spheres share a common centre $c \in \mathbb{R}$. By the functional equation (Theorem 1.1), if ρ is a nontrivial zero then so is $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$, whose sphere $S_{1-\bar{\rho}}$ is centred at $1 - \sigma_\rho$. Concentricity forces $\sigma_\rho = c = 1 - \sigma_\rho$, hence $\sigma_\rho = \frac{1}{2}$; as ρ was arbitrary this is RH, and the common centre is $\frac{1}{2}$. ☐

Part III
The Categorical Construction and the Concentricity Theorem

Part 2 revealed that ζ carries a special geometry: spread onto the octonions, its zeros are concentric exactly when the Riemann Hypothesis holds. That raises the question this part answers. What general properties of a section of the ring $\mathcal{R}$ of slice-preserving functions force its residue-$\mathbb{C}$ zeros to be concentric? Following that question leads to four conditions, C1–C4, and to the whole class of functions satisfying them, which we call the A-sections. The main result, the Concentricity Theorem, is that the residue-$\mathbb{C}$ zeros of any A-section are concentric. The octonionic zeta is one A-section among many, and the Riemann Hypothesis is one of its corollaries.

From the section's own data we read one distinguished exponential Möbius action on the Riemann sphere, presented by the Euler product on the half-space and by the Weierstrass product at N; orbit–stabilizer positions that single action at every object of the projective base. The resulting map to SphereWorld is its geometric projection, while the same action on normalized A-value states assembles the functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. Its Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ gathers every state and every value-transport into one total object. The zero-sphere system is the inverse-image subdiagram $\mathcal{R}_A$, included by $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$; the relative orbit–stabilizer comparison makes $\int_{\mathcal{B}} \mathcal{R}_A$ transitive and hence connected. The component colimit therefore collapses to a singleton, and its real-value readout is one centre $c \in \mathbb{R}$. The concentricity proof takes place in this residue-$\mathbb{C}$ inverse-image system inside the ambient total.

9 The geometric worlds and the categorical construction

This section assembles the parts the detector runs on. Part 2 established that exponentials are slice-preserving and live in the ring $\mathcal{R}$; this section develops a theory for working with these functions, one that shows how they *act*, from the point of view of action-groupoid theory. The key categories, the ones that capture the analysis and the geometry of these actions, are three: the octonionic world $\mathcal{H}_1$; the projective base $\mathcal{B}$, by the slice-preserving literature the home of the unique tame continuous lift of the degenerate GPV fibre; and a world of Riemann spheres, a whole continuum of

$\mathbb{C}_I^*$, each with its own Möbius self-maps. The general exponential slice-preserving action, and the A-section-specific action the proof uses, naturally carry value states of $\mathbb{O}^*$ to value states of $\mathbb{O}^*$; the theory requires slice-preserving normalization, so a functor on $\mathbb{O}^*$, the A-section equivariant functor below, assembles what such a total value system looks like: the total Grothendieck construction. To get there, we first explain the theory minimally, for one exponential action: orbit stabilizer makes the main slice-preserving functor well defined, and the one construction is simultaneously a value state, a group element, an action, and a functor. The same device defines the functors used in the total Grothendieck construction and establishes their properties, transitivity of the A-section's C-residue subsystem among them.

The categorical inventory is small, and we lay it out once. Five standing groupoids are introduced: the octonionic world $\mathcal{H}_1$, the sphere world, the projective base $\mathcal{B}$, the total $\mathcal{T}_A$, and the C-residue total $\int_{\mathcal{B}} \mathcal{R}_A$, with the G_2 fibre groupoids $\mathcal{A}_A(b)$ and $\mathcal{R}_A(b)$ standing over the base inside the two totals. Three functors run between them: the equivariant realization $A_{\mathbb{O}}$, the slice functor A^{slice}, and the A-section functor $\mathcal{A}_A$. Each depends on the disk action, positioned by orbit stabilizer, so the paper first explains how a slice-preserving functor is built at all. There is a general theory here, slice-preserving ring action groupoids over $\mathbb{O}^*$, one for every element of $\mathcal{R}$; this paper builds the A-section instance, because the hypotheses select one element. The point of building functors is the tools they unlock: the total Grothendieck construction is what this proof uses, and other tools of action-groupoid theory and categorical homotopy stand ready. Thomason's theorem [25] supplies an alternative proof route to the concentricity theorem formalized here; we do not take it. The proof itself then runs in one line: the hypotheses assemble the A-section-specific slice-preserving total, its inverse C-residue system is transitive, transitivity makes it connected, connectedness makes π_0 a singleton, and two readout theorems, running through the functors $\int_{\mathcal{B}} \mathcal{R}_A$ and π_0, turn the singleton into the one real number.

What would the graph of a slice-preserving section look like? For an ordinary function one draws the set of pairs (input, value). Here input and value both live on the compactified octonions, so the graph sits in sixteen real dimensions, and no one will ever see it. But slice preservation organizes it completely: the value at a point lies on that point's own slice sphere, and one holomorphic stem is repeated in every direction. The graph is one object in the octonions, and each direction $I \in S^6$ shows it as an ordinary graph: the slice Riemann sphere $\mathbb{C}_I^*$ with the pairing (input coordinate, value coordinate) of the restricted stem, the same picture in every direction, glued along the shared real circle and the shared point N. The construction of this section is that description made categorical, and it is the situation the Grothendieck construction was made for: since the fibered categories of SGA1 it has held one total object as compatibly glued fibres over a base. A state will record an input coordinate together with its positioned image and the value realized there, a point of the graph made into an object, and the groupoids will record the symmetries that carry graph points to graph points: projective transformations of the shared real circle, Möbius motions within each sphere, and G_2 rotations of the direction. A reader who keeps "one graph in the octonions, seen one slice sphere at a time" in mind has the right picture for everything below.

The compactified octonions and the point at infinity

Definition 9.1 (The compactified octonions $\mathbb{O}^* = S^8$)**.** Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$; the slice Riemann-sphere structure on S^m, $m = \dim_{\mathbb{R}} K \in \{4, 8\}$, is GPV [17], Prop. 4.1 and Thm. 4.2. The adjoined point is the north pole N; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.-Compactification.OnePoint`, $\mathbb{O}^* =$ `OnePoint` $\mathbb{O}$ with $\infty =$ `OnePoint.infty`, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.-`

`Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$, the point at infinity $\infty = N$. The groupoids $\mathcal{H}_1$ and SphereWorld are built on these compactified values (Definition 9.3); the distinct projective base $\mathcal{B}$ and direction-indexed sphere world used in the construction below are introduced in Definition 9.23.

There is one geometric north pole $N = \infty$, and it plays a role in each groupoid. In $\mathcal{H}_1$ this point of $\mathbb{O}^*$ is a G_2-fixed object: its own connected component, with $\mathrm{Aut}(N) = G_2$, carrying no universal property. In SphereWorld it is the common point at infinity of the slice Riemann spheres, fixed by every direction arrow. The section relates the two roles through C1: the pole of A has value ∞, so the section carries the pole point to N on every slice sphere at once. The residue field of a closed zero sorts the picture: residue-$\mathbb{R}$ (the G_2-fixed locus $\mathbb{R} \cup \{N\}$, classically trivial) versus residue-$\mathbb{C}$ (the 6-sphere orbits, classically nontrivial; the classification is Part 2 and Theorem 8.2). The later total construction uses this one shared compactified point N, the same for every zero index.

Looking forward: everything the construction does at N happens in action groupoids: N is an object, and the arrows are given by group elements, the distinguished Möbius action and G_2. N serves as a base object, as the shared point of every slice sphere, and as the value of the section at its pole; the section's own value at the domain point N enters as the marked datum of Remark 1.7. The hypotheses C1–C4 of Definition 10.4 are stated at finite points, and the infinite Euler family of C2 with the infinite divisor of C3–C4 makes every A-section essentially singular at the domain point N; what Part 3 reads at N is what the construction supplies there — the geometry of the point and the marked value.

What a slice-preserving section does

Definition 9.2 (The section's slice data and equivariance)**.** Let $A \in \mathcal{R}$. By the I-independent lift (Definition 6.4; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$-valued components,

$$A(x + Iy) = F_1(x + iy) + I\,F_2(x + iy), \qquad F_1, F_2 \ \mathbb{R}\text{-valued},$$

the same for every $I \in S^6$. Three consequences used below:

(i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I: the value at a point lies on that point's own slice sphere (Definition 5.1).

(ii) *G_2-equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I)\,F_2 = g \cdot \big(F_1 + I\,F_2\big) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2-fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i). (For $\zeta_{\mathbb{O}^*}$ this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

(iii) *Blindness to the sphere direction.* On a G_2-orbit $S_{(\sigma,\gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

A slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2-morphisms already present in both worlds.

The octonionic slice-preserving action groupoids

Three groups act on this picture, and they are the only three. $PGL(2,\mathbb{R})$ acts on the compactified real axis all slices share — projective transformations of the one great circle. $G_2 = \mathrm{Aut}(\mathbb{O})$ acts on the directions: it fixes the real axis pointwise and rotates S^6 transitively. And each slice Riemann sphere carries the Möbius transformations of its own compactified plane. A group action is naturally a groupoid, and this is the only categorical device the paper needs: the *translation groupoid* of $G \curvearrowright X$ has the points of X as objects and the elements $g \in G$ satisfying $g \cdot x = y$ as its morphisms $x \to y$, so an arrow *is* one group element g with $g \cdot x = y$, and $\pi_0(G \ltimes X) = X/G$ recovers the orbits (Quillen [22, §1]; in Lean, `CategoryTheory.ActionCategory`). Every carrier below is a translation groupoid of one of these three actions, with a single deliberate exception: the sphere world SphereWorld is not one action groupoid but a *continuum* of them — one Riemann sphere per direction, each carrying its own Möbius action, with the G_2 arrows relabelling which sphere one stands on. The exception is deliberate: SphereWorld holds a continuum of Riemann-sphere pictures side by side, one for every direction — the multiplied viewpoints of the epigraph, and the form in which an eight-dimensional graph lets itself be seen. Concretely, the projective action will be read in this direction-indexed sphere-world groupoid: its objects are directions $I \in S^6$ equipped with their compactified slice spheres, its morphisms carry one G_2 automorphism and one Möbius transformation, and a normalized value state pairs such a direction with a point of its slice.

Definition 9.3 (The octonionic world and the slice decomposition)**.** The octonionic world is

$$\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*.$$

Its components are the G_2-orbits: the compactified real fixed locus $\mathbb{R} \cup \{N\}$ and, for each $\sigma \in \mathbb{R}$ and $\gamma > 0$, the residue-$\mathbb{C}$ sphere $S_{(\sigma,\gamma)} = \{\sigma + \gamma v : v \in S^6\}$, labelled by its real centre σ and radius γ (Baez [3]; Theorem 7.5).

SphereWorld organizes the same compactified values by their slice decomposition. Every slice contains the real axis, and

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \qquad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

Its objects are the directions $I \in S^6$ with their compactified slice spheres — one sphere per direction, each with the Möbius action of its own compactified plane, the G_2 arrows relabelling which sphere one stands on; the points 0 and N lie on every slice at once. Möbius transformations thus enter the construction twice: complex ones as each slice sphere's own action in SphereWorld, and real ones as the projective base below, $\mathrm{PGL}(2,\mathbb{R})$ acting on the one compactified real line all slices share.

Slice preservation itself determines what must be tracked: the projective great circle every slice shares — the base $\mathcal{B}$ below; the continuum of Riemann spheres — SphereWorld; and the octonionic world $\mathcal{H}_1$, where the image of a slice-preserving function assembles them.

The projective base

The base is the projective action groupoid

$$\mathcal{B} = PGL(2,\mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R}).$$

Its objects are points of the compactified real projective line, the real great circle of $\mathbb{O}^*$: the slice through the real axis that every slice sphere shares. Its arrows are projective elements: an arrow $h : x \to y$ is the class of an invertible real matrix acting on the compactified line,

$$h = \left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right], \qquad h \cdot x = \frac{ax+b}{cx+d} = y,$$

one element of $\mathrm{PGL}(2,\mathbb{R})$ carrying x to y. We call these projective elements throughout.

The sphere world

The sphere world stands beside the base (`SphereWorld`), and it is the one category of this paper with a *continuum* of groups. Its objects are a continuum of Riemann spheres inside $\mathbb{O}^*$: one compactified slice sphere $\mathbb{C}_I^* \subset \mathbb{O}^*$ for each unit imaginary direction $I \in S^6 \subset \mathbb{O}$, each carrying the Möbius group of its own compactified plane. An arrow is a pair: one automorphism in G_2, relabelling which sphere one stands on, and one Möbius transformation, acting on the sphere; a normalized value state pairs a direction with a point of its slice. Beside the base the picture is complete: the base is the real Möbius world of the one line all slices share, and the sphere world is the continuum of complex Möbius worlds, one for each unit imaginary direction.

The disk action: a general slice-preserving action from the base to the sphere world

The aim is to understand slice-preserving maps on $\mathbb{O}^*$ in terms of action groupoids, and the two groupoids are already on the table: the projective base $\mathcal{B}$ just defined, whose arrows are classes of real matrices, and the sphere world just introduced, whose spheres each carry their own complex Möbius group. This subsection builds the generic material of an assignment from the first to the second: one complex Möbius element, produced by exponentiation and read onto every slice sphere by conjugation — a slice-preserving action. Everything here is general, with no A-section hypothesis used; the hypotheses are stated in the next subsection, and they specialize this general functor to the A-section's slice functor A^{slice} below. The object being built is a functor from $\mathcal{B}$ to the sphere world, and the values of a section already live in the sphere world: each value sits on its point's own slice sphere. Two groups are already chosen, and they are the only two. $\mathrm{PGL}(2,\mathbb{R})$ is the classes of invertible real matrices: the arrows of the base, acting on the compactified real line that every slice shares. $\mathrm{GL}(2,\mathbb{C})$ is the invertible complex matrices: its classes are the Möbius transformations, one group on each slice sphere, acting on that sphere's compactified plane. The real classes are already the real part of each Möbius group, and the disk automorphisms, the Blaschke maps, are their Cayley conjugates. This is what lets the main general slice-preserving exponential functor be built: not yet the A-section, and not an arbitrary element either, but a general *analytic* ring element, the kind we want, since it carries its degenerate GPV base. The construction uses two matrices, the Cayley matrix and the exponential:

$$C = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}, \qquad M(z) = \begin{pmatrix} e^z & 0 \\ 0 & 1 \end{pmatrix}:$$

C is the map $w \mapsto (w-i)/(w+i)$, carrying the base's compactified real line onto the unit circle of a slice sphere; M takes a complex number z to the diagonal matrix whose entry is e^z, exponentiation in matrix form. An invertible matrix acts on the Riemann sphere by

$$h = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2,\mathbb{C}), \qquad h \cdot w = \frac{aw+b}{cw+d}.$$

The class $[h]$ of the matrix up to scalars acts, and this is the Möbius map itself: a scalar multiple, all four entries scaled at once, enters numerator and denominator together and cancels,

$$\lambda h = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}, \qquad (\lambda h) \cdot w = \frac{\lambda a\, w + \lambda b}{\lambda c\, w + \lambda d} = \frac{\lambda\,(aw + b)}{\lambda\,(cw + d)} = h \cdot w, \qquad \lambda \in \mathbb{C}^{\times}.$$

The action is total on the compactified plane: numerator and denominator never vanish at the same w, for if both did, then $a(cw+d) - c(aw+b) = ad - bc$, the determinant, would vanish; where the denominator alone vanishes the value is the point at infinity N.

The functor being built must act on any slice sphere, at any normalization, so the first fact to establish is that the Möbius action reaches every point of a sphere.

Lemma 9.4 (The Möbius action is transitive)**.** *Any two points of a slice Riemann sphere are joined by one Möbius transformation: the action of the classes of* $\mathrm{GL}(2, \mathbb{C})$ *on the compactified plane is transitive.*

Proof. The translation to t and the inversion are the classes

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \cdot w = w + t, \qquad s = \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right], \quad s \cdot w = \frac{1}{w}.$$

The translations carry 0 to every finite value. The inversion carries 0 to N: at $w = 0$ its denominator alone vanishes. The action at N itself is computed through the group law, composing with s first:

$$h \cdot N = (hs) \cdot 0 = \begin{pmatrix} b & a \\ d & c \end{pmatrix} \cdot 0 = \frac{a}{c},$$

the point N itself when $c = 0$. So the action is transitive: any two points of the sphere are joined by one class, a composite of the displayed translations and inversion. $\square$

The diagonal matrix acts by

$$M(z) \cdot w = e^{z} w = e^{x}\big(e^{iy} w\big), \qquad z = x + iy,$$

the modulus scaled by e^{x} and the argument advanced by y.

Now the orbit–stabilizer construction. Take a value w somewhere on a slice Riemann sphere; by the transitivity just shown it lies in the orbit of the Möbius action, and in particular C is itself an invertible class, so there is exactly one point w_0 with

$$w = C \cdot w_0, \qquad w_0 = C^{-1} \cdot w,$$

the value standing as the C-positioned image of w_0. The matrix $M(z)$ acts at w_0 by $M(z) \cdot w_0 = e^{z} w_0$; to act on the value w where it stands, conjugate, and the group law computes the conjugated element doing exactly that:

$$(C\, M(z)\, C^{-1}) \cdot w = (C\, M(z)\, C^{-1}) \cdot (C \cdot w_0) = C \cdot \big(M(z) \cdot w_0\big),$$

the inner $C^{-1}C$ cancelling by the action law. This is orbit stabilizer, and it is the positioning pattern of the whole paper: to act at a positioned location, conjugate the action by the element that does the positioning — it runs over the base below at every object and arrow. The operation just performed gets the name it keeps.

Definition 9.5 (The Cayley conjugation and the disk action)**.** For any invertible complex matrix,

$$\mathrm{cayleyProjective}(h) = \big[C\,h\,C^{-1}\big], \qquad h \in \mathrm{GL}(2,\mathbb{C}),$$

and the passage from a complex number to a slice Möbius map is two steps, exponentiate then conjugate:

$$z \;\longmapsto\; M(z) \;\longmapsto\; D(z) = \mathrm{cayleyProjective}\big(M(z)\big) \in \mathrm{M\ddot{o}b}(\mathbb{C}^*).$$

cayleyProjective takes any invertible matrix, because the positioned real matrices of the base pass through the same map at the orbit–stabilizer factorization below (Lemma 9.15).

Lemma 9.6 (Homomorphism laws)**.** cayleyProjective *is a group homomorphism, and*

$$M(z+w) = M(z)M(w), \qquad D(z+w) = D(z)D(w).$$

Proof. The homomorphism is the same cancellation that ran the conjugation:

$$C\,(h_1 h_2)\,C^{-1} = C\,h_1\,(C^{-1}C)\,h_2\,C^{-1} = (C\,h_1\,C^{-1})(C\,h_2\,C^{-1}).$$

Matrix multiplication with $e^{z+w} = e^z e^w$ gives $M(z+w) = M(z)M(w)$, and applying the homomorphism cayleyProjective to that equation gives $D(z+w) = D(z)D(w)$. □

Every later appearance of cayleyProjective, positioning the distinguished element over the base and acting by a stabilizer factor, is this same map applied to a different matrix. Thus exponentiation is the homomorphism turning addition of logarithmic lifts into composition of the actual slice Möbius maps — recorded once, because every GPV lift below inhabits it:

$$\begin{tikzcd}
\Gamma(t) \arrow[r, "\exp"] \arrow[d, "D"'] & v(t) \in \mathbb{C}^\times \arrow[d, "u \mapsto \mathrm{cayleyProjective}(\mathrm{diag}(u,1))"] \\
D(\Gamma(t)) \arrow[r, equal] & \mathrm{cayleyProjective}\big(\mathrm{diag}(v(t),1)\big)
\end{tikzcd}$$

The top row is the logarithmic lift; the bottom row is the Möbius map it generates; the square commutes at every instant, and adding a winding $2\pi ik$ moves only the top row, $e^{\Gamma+2\pi ik} = e^{\Gamma}$. The Euler product will arrive as one completed logarithmic sum, and this square is how one sum acts as one group element at every instant. In the Lean source these are the chain `diagonalGL`, `diagonalGLHom`, `expUnit`, and `diskExpAction`, with their multiplication laws.

Slice preservation is itself a group action, and already one exponentiation carries its GPV degenerate base. For $u \in \mathbb{C}^\times$ the logarithms of u form the exponential fibre

$$\exp^{-1}(u) = \{\lambda + 2\pi ik :\; k \in \mathbb{Z}\}, \qquad M(\lambda + 2\pi ik) = M(\lambda),$$

the branches carried by the logarithm manifold of Theorem 6.2 (the commuting triangle $\pi \circ E = \exp$): a lift lives in this fibre, the winding is the integer k separating its elements, and M carries the whole fibre to the one matrix.

Exponentiation also resolves a puzzle the Euler product will pose. An infinite product of maps cannot enter a group as a composition: a composition has a last factor, and an infinite one does not. The logarithm is the way through. In the logarithmic coordinate an infinite product is an infinite *sum*, and sums, unlike compositions, complete: local normal convergence makes the completed sum a single complex number at every point where it is stated. (*Locally normally* means: every point has a neighbourhood on which the sum of suprema $\sum_p \sup|\ell_p|$ converges. This is the sense in which the cited Euler and Weierstrass products converge, Theorem 1.2 and Proposition 10.1; it gives uniform convergence on compacts, hence continuity of the completed sum, and it is carried in the development as the summability and majorant hypotheses of the A-section interface, proved outright for $\zeta_{\mathbb{O}^*}$ — `zetaC2_summable` and companions.) Exponentiation acts once, on the completed sum.

The slice facts of Part 2 make this passage well defined, and they act in order. First the commuting triangle of Theorem 6.2, the GPV logarithm manifold: the $E^+_{\mathbb{O}}$-exponential E and the first-factor projection π close over the exponential,

$$\begin{tikzcd} \mathbb{O} \arrow[r, "E"] \arrow[dr, "\exp"'] & E^+_{\mathbb{O}} \arrow[d, "\pi"] \\ & \mathbb{O}\setminus\{0\}, \end{tikzcd} \qquad \pi \circ E = \exp,$$

and the blow-up unwraps the multivalued logarithm once and for all: the fibre of π over a value v is the exponential fibre $\exp^{-1}(v)$, its elements the branches, carried into the manifold by E. That fibre is degenerate in its real coordinate, for its elements differ only by $2\pi ik$, purely imaginary, so the whole fibre over v stands on the one real number $\log\|v\|$: the unique GPV real fibre, taken up in detail when the A-section functors are built below. Along a path δ, a continuation of the logarithm is a lift Γ with $e^{\Gamma(t)} = v(t)$; it exists whenever the value stays away from zero along the path, and tameness makes it unique once its initial value $\Gamma(0)$ is fixed in the exponential fibre (Theorem 6.5, Proposition 6.6). The winding is the integer: two lifts of the same value path differ by one constant $2\pi ik$, and exponentiation is blind to it, $e^{\Gamma+2\pi ik} = e^{\Gamma}$, so every branch generates the same matrix at every instant while the real part $\operatorname{Re}\Gamma(t) = \log\|v(t)\|$ is the one real value fixed by the degenerate fibre. This is the general well-definedness: for any slice-preserving section and any pole-avoiding, zero-free path, the completed logarithmic data determines one group element and one real value at each instant, independent of every choice made along the way.

The A-section hypotheses

Here the hypotheses enter — their first working appearance, so we state them informally at once; the formal statement is Definition 10.4. An *A-section* is a section $A \in \mathcal{R}$ with: (C1) a meromorphic continuation with exactly one pole — simple, real, its value the point at infinity N; (C2) an infinite prime-indexed Euler product on a right half-space, each term a power series in p^{-q} for its own prime — one logarithmic base; (C3) an infinite slice-regular Weierstrass factorization whose sphere factors enumerate the residue-$\mathbb{C}$ zeros; (C4) infinitely many such zeros. Two consequences orient everything that follows. First, the hypotheses manufacture one number: C1 continues the pole-cancelled factor through the simple pole, C2 and C3 present that same factor on a punctured neighbourhood of the pole, and its value at the pole is the multiplier $u_A \neq 0$ — the infinite structure completed and evaluated, which M turns into one matrix. Second, the hypotheses place a marked point at N (Remark 1.7): a section with infinitely many zeros is not rational, so its domain point N carries an essential singularity, and the compactification supplies what the construction uses there — the geometry of the point and the values positioned at it. Every statement this paper makes at N is of that kind.

The general machine now specializes to what the hypotheses supply. Shown at the A-section itself, on the zero-free half-space of C2:

$$\Gamma_A(z) = \sum_p \ell_p(z), \qquad e^{\Gamma_A(z)} = A(z), \qquad M\big(\Gamma_A(z)\big) = \operatorname{diag}\big(A(z), 1\big) :$$

the infinite prime-power degenerate base completes to one sum, exponentiation acts on it once — $M(z+w) = M(z)M(w)$ is the homomorphism from addition to composition — and the infinite family enters the group as one matrix. This is also why the half-space is zero-free: on it $A = e^{\Gamma_A}$ is a value of the exponential, and the exponential is nowhere zero (Theorem 6.1). The unique tame lift re-shows the same construction along any path: a continuation Γ with $e^{\Gamma(t)} = A(\delta(t))$ and fixed initial value is unique (Theorem 6.5), so $M(\Gamma(t))$ is the matrix of the value at every instant, the same for every branch of the fibre.

The slice projection $A^{\mathrm{slice}} : \mathcal{B} \to \mathrm{SphereWorld}$

Conditions C1–C3 construct one element, and this subsection builds it and the slice functor it defines.

Definition 9.7 (The multiplier and the distinguished disk action D_A)**.** Write p_A for the pole of A and $g_A =$ `distinguishedPoleFactor A` for the pole-cancelled factor. On a punctured neighbourhood of the pole, conditions C2 and C3 present this one factor twice:

$$g_A(z) = (z - p_A)\exp\left(\sum_p \ell_p(z)\right),$$

$$g_A(z) = z^{m_A} R_{\mathrm{fac},A}(z) e^{h_A(z)} \prod_{n\geq 0} \mathcal{E}\big(z; q_{A,n}\big).$$

Condition C1 continues this same factor through the simple pole — the finite point p_A, never the domain point ∞ (Remark 1.7); the sum diverges at p_A itself, the factor does not — and evaluates it there:

$$u_A = g_A(p_A) \neq 0.$$

This is the one number the hypotheses manufacture: the infinite structure completed and evaluated. Its logarithms form the exponential fibre of the previous subsection, $\exp^{-1}(u_A) = \{\lambda_A + 2\pi i k : \ k \in \mathbb{Z}\}$, and the function/group-element passage is the commuting square

$$\begin{tikzcd}
z \arrow[r, "\exp"] \arrow[d, mapsto] & e^z \in \mathbb{C}^\times \arrow[d, mapsto] \\
M(z) = \operatorname{diag}(e^z, 1) \arrow[r, "\text{cayleyProjective}"'] & D(z) \in \text{Möb}(\mathbb{C}^*).
\end{tikzcd}$$

through which M collapses the fibre to the one matrix

$$M(\lambda_A) = \operatorname{diag}\big(e^{\lambda_A}, 1\big) = \operatorname{diag}(u_A, 1),$$

so the element the hypotheses construct is

$$D_A = D(\lambda_A) = \text{cayleyProjective}\big(\operatorname{diag}(u_A, 1)\big).$$

Here λ_A is the logarithm and M is what undoes it: the entry of $M(\lambda_A)$ is $e^{\lambda_A} = u_A$, the multiplier itself, so D_A is an exponential action and not a logarithmic one. The logarithm is the coordinate in which the lift moves; the element it generates lives one exponential above it. Every element of the fibre gives this same matrix and hence this same D_A: the logarithm carries the winding, and the exponential action it generates is one element. This is what lets the lift run through the whole base while D_A stays one element. The Euler and Weierstrass expressions displayed above are the two presentations of this same factor.

The element constructed, the slice functor now stands on it.

Definition 9.8 (The slice functor A^{slice})**.** The direction-and-Möbius projection is

$$A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld},$$

written `AsectionSlice A` in the Lean source. It is a functor, and its two assignments are built from three constructed pieces: the projective action on the base is transitive, so one projective element o_b is fixed for each object, $o_b \cdot N = b$, and every arrow h from a to b factors through the north stabilizer, $h = o_b\, r_h\, o_a^{-1}$; the disk action is D_A; and cayleyProjective carries the base's matrices onto the slice spheres. On objects: b goes to the direction-indexed slice spheres on which D_A acts through the representative,

$$\text{cayleyProjective}(o_b)\, D_A.$$

On arrows: h goes to the sphere-world arrow whose G_2 automorphism is the identity and whose Möbius transformation is the factorization read through the same builds,

$$\big(\text{cayleyProjective}(o_b)\, D_A\big)\ \text{cayleyProjective}(r_h)\ \big(\text{cayleyProjective}(o_a)\, D_A\big)^{-1}.$$

The representatives o_b and the factorization with its uniqueness and laws are Definition 9.14 and Lemma 9.15 below. What makes the two assignments well defined is D_A: it is the one factor either formula takes from the A-section, and every branch of the exponential fibre gives that same element (Definition 9.7). The functor laws are proved at Proposition 9.17 below, read through the disk action: $D(z+w) = D(z)D(w)$ turns sums into compositions, and cayleyProjective carries $r_{1_a} = 1$ and $r_{h_1 ⨟ h_2} = r_{h_2} r_{h_1}$ onto the Möbius transformations ($⨟$ is composition in diagrammatic order, the left factor first); equal completed data gives equal transformations.

A^{slice} records the spheres and the positioned Möbius motion, and the equivariant realization $A_{\mathbb{O}}$ below realizes the section's value on them. The full A-section functor below retains, in addition, the normalized point and the value produced on that sphere.

Exponentiating the prime-power sum — the A-section action, which becomes Weierstrass at N — is why there is a fixed real-value transport: the action consumes the path from $\delta(0)$ to $\delta(1)$ as matrix group actions, $D(\Gamma(t))$ at every instant, hence a real value transported. The GPV transport of the next subsection records exactly this.

The GPV transport

The GPV lift records how g_A varies between its presentations.

Definition 9.9 (The GPV transport)**.** Let $a, b \in \mathcal{B}$. A projective-base transport from a to b carries three synchronized data: a domain path δ^*, recording where the section is read; a value v along it; and a logarithmic lift Γ, the continuous logarithm of the value. Each is a continuous function of one parameter $t \in [0,1]$, the instant of the transport. The instant parametrizes the transport from a to b and belongs to the transport alone; the geometry keeps its own coordinates, and the three data are read together at each instant. The hypotheses are what make the instant necessary. C1 is a continuation, and continuation happens along a path: the one factor is continued to the pole, and its value at each instant is the continued value. The logarithm of C2's completed sum becomes single-valued along a path: the unique tame lift fixes every branch from the initial one, which is the GPV mechanism itself. And the argument reads what the path carries: the one factor from its Euler presentation on the half-space to its Weierstrass presentation at N, one real value carried the whole way. What varies with the instant is the prime-power sum: the degenerate base of C2 is evaluated along the domain path, and the matrix it generates varies with it. What the matrix acts on does not vary: an input, one fixed point of its locus, stands still. The endpoints of the path are determined by the hypotheses: it begins in the Euler half-space of C2, where the prime-power degenerate base converges, and ends at the pole of C1, for the octonionic zeta the pole at $s = 1$; the instants $t = 0$ and $t = 1$ are this beginning and this arrival. Here the critical strip's classical mystery takes its present form: the strip lies beyond the half-space where the prime-power base converges, and the zeros stand there as fixed coordinates; the path is how the arithmetic of the primes reaches them, as the matrix produced at the pole. The value at the pole generates that matrix, and the matrix acts on the standing input: the prime-power sum varies in the construction of the matrix, and the input it acts on is fixed. The section takes its values on the compactified sphere, and the exponential fibres live one exponential below, over $\mathbb{C}^\times$; so the value is carried twice at once, as the finite nonzero $v(t) \in \mathbb{C}^\times$ that the exponential reaches, and as its reading $v^*(t)$ on the compactified sphere, the same value seen where the section takes it. Their defining equations are

$$\begin{aligned} \delta^*(0) &= \mathrm{complexPoint}(a), & \delta^*(1) &= \mathrm{complexPoint}(b), \\ v^*(t) &= A\big(\delta^*(t)\big), & e^{\Gamma(t)} &= v(t), \\ \Gamma(1) - \Gamma(0) &= 2\pi i k, & k &\in \mathbb{Z} : \end{aligned}$$

the domain path runs from a to b; the value, read on the sphere, is the section's value along it; the lift exponentiates to the value where the exponential lives; and the winding field records the circuit case. These are precisely the fields of `ASection.GpvTransport`.

At each t the lift equation passes through the matrix square:

$$\begin{tikzcd} \Gamma(t) \arrow[rr, "\exp"] \arrow[d, "D"'] & & v(t) \in \mathbb{C}^\times \arrow[d, "u \mapsto \mathrm{cayleyProjective}(\mathrm{diag}(u,1))"] \\ D(\Gamma(t)) \arrow[r, equal] & \mathrm{cayleyProjective}\big(\mathrm{diag}(v(t), 1)\big). & \end{tikzcd}$$

Thus every point of the lift is already an action of the same kind as D_A.

Lemma 9.10 (The real level)**.** *Along a GPV transport the real level* $\operatorname{Re}\Gamma(t) = \log\|v(t)\|$ *is continuous, depends only on the value along the path, and is the same for every lift of that value path.*

Proof. The lift equation is $e^{\Gamma(t)} = v(t)$. Take norms: the slice exponential has $\|e^{\Gamma(t)}\| = e^{\operatorname{Re}\Gamma(t)}$ (Theorem 6.1), so

$$e^{\operatorname{Re}\Gamma(t)} = \|v(t)\|, \qquad \operatorname{Re}\Gamma(t) = \log\|v(t)\| :$$

the real level is continuous and depends only on the value along the path. Uniqueness and branch-independence are the cited GPV facts (Theorem 6.5, Proposition 6.6): the tame lift is unique once its initial value $\Gamma(0)$ is fixed, and any two lifts of the same value path differ by one purely imaginary constant $2\pi i k$, so every lift carries the same real part ([21, Cor. 4.13]). □

At each instant the action is $D(\Gamma(t))$, moving with the value along the domain path: one group element and one real value at every instant, the branch ambiguity spent fibrewise on the degenerate fibre. The winding field records the circuit case, the path whose value returns, and its integer counts the branches crossed. This is the fixed real-value transport: the action consumes the path from $\delta(0)$ to $\delta(1)$ as matrix group actions, and the real value is transported.

Condition C2 now supplies the canonical arithmetic instance of this construction. Along an Euler half-space loop δ,

$$\Gamma_A(t) = \sum_p \ell_p(\delta(t)), \qquad e^{\Gamma_A(t)} = A(\delta(t)),$$

with the prime index retained inside the complete summation. Local normal convergence makes Γ_A continuous, the zero-free half-space makes its value lie in $\mathbb{C}^\times$, and the GPV uniqueness theorem determines the whole lift from its initial value $\Gamma_A(0)$ in the exponential fibre. The sum is complete at every instant; only the evaluation point moves along the domain path. Since $\delta(0) = \delta(1)$, the completed sum is evaluated at the same point at both ends, so the checked winding is $k = 0$: the loop returns to the same element of the fibre, $\Gamma_A(1) = \Gamma_A(0)$, level and matrix with it.

The same construction runs to the one north object.

Lemma 9.11 (Arrival at the north object)**.** *Run the domain path of a GPV transport of the continued factor to $\delta(1) = p_A$. Then*

$$v(1) = g_A(p_A) = u_A, \qquad D(\Gamma(1)) = D(\lambda_A) = D_A, \qquad \operatorname{Re}\Gamma(1) = \operatorname{Re}\lambda_A,$$

and two runs arriving at N differ by one whole winding, $\Gamma_2(1) - \Gamma_1(1) = 2\pi i k$ with $k \in \mathbb{Z}$.

Proof. The value along the run is $g_A(\delta(t))$, nonzero through the pole, entering from the Euler half-space where $g_A = (z - p_A)\, e^{\Gamma_A}$ and arriving at $v(1) = g_A(p_A) = u_A$, where condition C3 presents the multiplier u_A in Weierstrass form. The arriving action is D_A as Definition 9.7 constructed it, $D_A = D(\lambda_A) = \mathrm{cayleyProjective}\big(\mathrm{diag}(u_A, 1)\big)$, and its level is $\operatorname{Re}\lambda_A$ by Lemma 9.10. Every run arrives at this same element and the same real value, and two lifts arriving at N differ by one purely imaginary constant $2\pi i k$, the whole winding. In the development these are `ASection.RegularizedNorthRun` with `diskExpAction_at_pole`, `level_at_pole`, and `joint_winding`; the square `canonicalAsectionPresentation_euler_toNorth` carries the complete prime-power lift to the one north object at every instant. □

The prime sum, its value under A, its logarithmic level and winding, and the matrix action it generates are the data of one element, $D(\Gamma(t))$ at each instant, travelling together.

Condition C4, infinitely many residue-$\mathbb{C}$ zeros, matches the class to its infinite structure — the infinite Euler family of C2, the infinite Weierstrass divisor of C3 — and populates the degenerate fibre of the transport with an infinite family; Theorem 10.8 reads one real value across the whole population.

The equivariant realization $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$

An A-section is a ring element: one map from $\mathbb{O}^*$ to $\mathbb{O}^*$, its value at each point standing on that point's own slice sphere. The slice functor just built records where the inputs are positioned; what the total Grothendieck construction needs beside it is a functor that records the *realized values*: the round trip, input in $\mathbb{O}^*$ and value in $\mathbb{O}^*$, carried by the same G_2 arrows. This subsection defines that functor.

Definition 9.12 (The equivariant realization)**.** The realization is a functor between two groupoids already built, from the octonionic world to itself:

$$A_{\mathbb{O}} : \mathcal{H}_1 \longrightarrow \mathcal{H}_1, \qquad \mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*,$$

written `AsectionEquivariant` in the Lean source. The values it records are produced by A^{slice}, just defined via the disk action — the A-section–specific action carried from $\mathcal{B}$ to the slice spheres,

$e^{\Gamma_A} = A$ on the zero-free half-space and the continued factor through the pole. On objects: a point q of $\mathbb{O}^*$ goes to the value produced there, $A^{\text{slice}}(q)$. On arrows: an arrow of $\mathcal{H}_1$ is one element $g \in G_2$ with $g \cdot q = q'$, and it goes to the same element,

$$A_{\mathbb{O}}\big(g : q \to q'\big) \;=\; g : A^{\text{slice}}(q) \to A^{\text{slice}}(q').$$

Proposition 9.13 ($A_{\mathbb{O}}$ is a functor)**.** *The two assignments of Definition 9.12 define a functor* $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$*.*

Proof. The target is an arrow of $\mathcal{H}_1$ because equivariance (Definition 9.2(ii)) gives $g \cdot A^{\text{slice}}(q) = A^{\text{slice}}(g \cdot q) = A^{\text{slice}}(q')$. The functor laws are the group's own: the identity $1 : q \to q$ goes to $1 : A^{\text{slice}}(q) \to A^{\text{slice}}(q)$, and a composite hg goes to hg, the same group elements on both sides. □

A^{slice} and $A_{\mathbb{O}}$ are two functors of the same slice-preserving action: the first records the projective direction-and-Möbius transport, the second the value of the section, retaining the same G_2 arrow by equivariance. Condition C2 identifies the element acting in both with the exponentiated log sum of prime powers — the Euler product — of the given A-section; condition C3 presents the same multiplier in Weierstrass form at N and selects the residue-$\mathbb{C}$ locus; and C4 makes the selected population infinite. The A-section functor of the next subsection consumes both.

The unique real fibre

The section opened with the graph made categorical: one total object held as compatibly glued fibres over a base, the situation the Grothendieck construction was made for since the fibered categories of SGA1. What makes the gluing work for an A-section is GPV uniqueness, and this subsection shows how.

Three facts are already on the table. The prime-power sum is what is exponentiated: $M(\Gamma_A)$ is one of the two matrices of A^{slice}, the Cayley matrix C the other. There is one N on $\mathbb{O}^*$: the single point at infinity, shared by every slice sphere. And the great circle base slices $\mathbb{O}^*$ through the real axis: the compactified real line every slice shares, the groupoid $\mathcal{B}$.

The group theory to show is that the projection to the fibre at b is unique.

Definition 9.14 (The orbit representatives)**.** For each point b of the base fix one projective element o_b with $o_b \cdot N = b$, once — the development's `GreatCircle.orbitRep` — with $o_N = 1$ at the north object.

Lemma 9.15 (North stabilizer uniqueness)**.** *Every arrow* $h : a \to b$ *of* $\mathcal{B}$ *factors through the north stabilizer* $\mathrm{Stab}(N) = \{r : r \cdot N = N\}$ *in exactly one way,*

$$h = o_b\, r_h\, o_a^{-1}, \qquad r_h = o_b^{-1} h\, o_a \in \mathrm{Stab}(N),$$

and the stabilizer factor obeys the two laws

$$r_{1_a} = 1, \qquad r_{h_1 \mathbin{⨾} h_2} = r_{h_2}\, r_{h_1}$$

for every composable pair $h_1 : a \to b$*,* $h_2 : b \to c$*.*

Proof. Take an arbitrary arrow of the base, $h : a \to b$: one element of $\mathrm{PGL}(2,\mathbb{R})$ with $h \cdot a = b$. Its source a has the fixed presentation $o_a \cdot N = a$, its target b has $o_b \cdot N = b$. Form the product $o_b^{-1} h\, o_a$ and let it act on N, one factor at a time, the right factor first:

$$(o_b^{-1} h\, o_a) \cdot N = o_b^{-1} \cdot \big(h \cdot (o_a \cdot N)\big) = o_b^{-1} \cdot (h \cdot a) = o_b^{-1} \cdot b = N.$$

The first equality is the action law; the second is the defining equation of o_a; the third is $h \cdot a = b$; and the last inverts the defining equation of o_b, from $o_b \cdot N = b$ to $o_b^{-1} \cdot b = N$. So $r_h = o_b^{-1} h\, o_a$ lies in $\mathrm{Stab}(N)$, and $h = o_b\, r_h\, o_a^{-1}$. The factor is unique: if $h = o_b\, r\, o_a^{-1}$ for any stabilizer element r, multiplying on the left by o_b^{-1} and on the right by o_a gives

$$r = o_b^{-1} h\, o_a = r_h.$$

For the identity law, the identity arrow $1_a : a \to a$ has both ends at a, so the formula $r_h = o_b^{-1} h\, o_a$ reads with o_a at both ends, and the representatives cancel:

$$r_{1_a} = o_a^{-1}\, 1\, o_a = 1.$$

For the composition law take a composable pair — the codomain of h_1 is the domain of h_2 — $h_1 : a \to b$ and $h_2 : b \to c$. Their composite $h_1 \mathbin{⨟} h_2 : a \to c$ is composition in diagrammatic order, h_1 first, then h_2, and as matrices it is the product $h_2 h_1$. Its ends are a and c, so the formula reads $r_{h_1 ⨟ h_2} = o_c^{-1}(h_2 h_1)\, o_a$, and inserting $o_b\, o_b^{-1} = 1$ between the two matrices splits it into the two stabilizer factors:

$$r_{h_1 ⨟ h_2} = o_c^{-1}(h_2 h_1)\, o_a = (o_c^{-1} h_2\, o_b)(o_b^{-1} h_1\, o_a) = r_{h_2}\, r_{h_1}.$$

□

The projection from N to each base point b is therefore the fixed o_b, and what an arbitrary arrow adds is exactly its stabilizer factor r_h.

Remark 9.16 (The unique GPV real fibre). The real fibre is what is exponentiated. Over each base object stands the unique GPV real fibre: the branches of the prime-power lift differ by $2\pi i k$, the whole fibre stands on one real number, and M carries every branch to the same matrix — the collapse $M(\lambda + 2\pi i k) = M(\lambda)$ of Definition 9.7. The fibre stands on one real number even though A^{slice} is built from an Euler product, because the sum completes: local normal convergence makes the prime-power sum one complex number at every point where it is stated, its exponential one value, and that value's logarithms one exponential fibre — the completed sum is one branch in it. At the pole p_A the completed structure arrives as the multiplier u_A, presented by C3 in Weierstrass form, and it still has a unique degenerate base: $\exp^{-1}(u_A) = \{\lambda_A + 2\pi i k : \ k \in \mathbb{Z}\}$, standing on the one real number $\mathrm{Re}\, \lambda_A$.

Proposition 9.17 (A^{slice} is a functor)**.** *The two assignments of the slice functor (Definition 9.8) — the positioned disk action at each object, the sphere-world arrow at each arrow — define a functor* $A^{\mathrm{slice}} : \mathcal{B} \to \mathrm{SphereWorld}$.

Proof. A^{slice} was defined above, and its definition deferred exactly this material — the representatives, the factorization, and the functor laws — to the present point. At an object b: the representative o_b is one element of $\mathrm{PGL}(2, \mathbb{R})$; cayleyProjective carries it onto the slice spheres, and the action at b is the disk action so carried,

$$\mathrm{cayleyProjective}(o_b)\, D_A;$$

at N, $o_N = 1$, and it is D_A itself. At an arrow $h : a \to b$: h acts by what it adds, its stabilizer factor r_h (Lemma 9.15), through the same conjugation shown at the disk action: to act on points placed by one element, conjugate by that element. Here points are placed by the actions at a and at b, so the Möbius transformation of h is

$$\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}. \tag{OS}$$

Read right to left: the inverse undoes the action at a, cayleyProjective(r_h) acts by what h adds, and the action at b places the result at the target — the inverse is there exactly to undo the source placement first. We refer to (OS) by name throughout.

For the functor laws, apply the homomorphism cayleyProjective — the same conjugation that produced D_A and the positions — to the two laws of Lemma 9.15. A homomorphism sends the identity to the identity and products to products, so the two equations hold among the Möbius transformations of the A-section's action:

$$\mathrm{cayleyProjective}(r_{1_a}) = 1, \qquad \mathrm{cayleyProjective}(r_{h_1 ⨾ h_2}) = \mathrm{cayleyProjective}(r_{h_2})\ \mathrm{cayleyProjective}(r_{h_1}).$$

At 1_a the arrow's Möbius transformation collapses to the identity, and at a composite it is the composite of the two arrows' transformations, the position at b cancelling between them as o_b did in the matrices. $\square$

The quick recap: one N on $\mathbb{O}^*$; one base, the great circle slicing through the real axis; one prime-power sum, exponentiated. The projection from N to each base point b is the fixed o_b, and every arrow h adds exactly its stabilizer factor r_h in $\mathrm{Stab}(N)$. The exponential fibre over each value is the set of its logarithms, the branches differing by $2\pi ik$; the completed prime-power sum is one branch in it, and by GPV uniqueness the whole fibre stands on one real number. Every branch exponentiates to the same A-specific disk action, at N the arrival D_A. The two laws of Lemma 9.15 make A^{slice} a functor (Proposition 9.17), and the comparison in Lemma 10.7 consumes these same facts.

The functor for the total Grothendieck construction: $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$

The total Grothendieck construction of the next subsection consumes one input: a functor from the base to **Grpd**, and every ingredient of this one is already built. An object of **Grpd** is a groupoid; $\mathcal{H}_1$, the sphere world, and $\mathcal{B}$ are three of its objects. An arrow of **Grpd** is a functor between groupoids, and the two functors just built are two of its arrows: $A^{\mathrm{slice}} : \mathcal{B} \to \mathrm{SphereWorld}$ records where the inputs are positioned, and $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$ records the values A^{slice} produces. The functor this subsection builds lands in **Grpd** itself:

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd}.$$

To each base object b it assigns a groupoid: its objects are the A-section's states at b, and its arrows are elements of the group G_2, one element γ with $\gamma \cdot x = y$ — the translation groupoid of G_2 acting on those states. To each projective element $h : a \to b$ it assigns the transport of those states: each state moves by the Möbius transformation of $A^{\mathrm{slice}}(h)$ (`projectiveArrowHom`, `projectiveArrowElement`). The states are positioned by the A-section's action, and the hypotheses built that action once, at the pole p_A: C1 continues g_A through the pole, the value u_A there generates the disk action D_A, and C3 gives it its Weierstrass form. The total construction glues these groupoids, one per point of the real great circle, into $\mathcal{T}_A$ of the next subsection. The source $\mathcal{B} = \mathrm{PGL}(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$ is the projective action groupoid: an object is a point of the compactified real projective line, and an arrow $h : a \to b$ is one element of $\mathrm{PGL}(2, \mathbb{R})$ with $h \cdot a = b$. The base is this circle for the A-section's own reason: the compactified real axis is the one line every slice shares, the pole p_A and the north object N are its points, and over each of its objects stands the complete prime-power GPV lift, determined by its initial value $\Gamma_A(0)$ in the exponential fibre, which the disk action consumes.

Each point of $\mathbb{O}^*$ off the real axis lies on exactly one slice sphere, so it is two data: the sphere, named by its direction $I \in S^6$, and the point on it, $q = x + Iy$. Its slice coordinate is the complex number $u = x + iy$, the point read on the one compactified plane $\mathbb{C}^*$ that every slice identifies with — normalized means read there, the same plane for every direction. The pair (I, u) is that point in components, $q = (I, u)$, paired as the normalized value states introduced with the sphere world, and $\gamma \cdot (I, u) = (\gamma I, u)$.

Definition 9.18 (The A-section states and their fibre groupoids)**.** On a base object b, the groupoid $\mathcal{A}_A(b)$ consists of the A-section states positioned at b by A^{slice}. A state is one triple,

$$\Big((I, u), \quad A^{\text{slice}}(b) \cdot (I, u), \quad A_{\mathbb{O}}\big(A^{\text{slice}}(b) \cdot (I, u)\big) \Big) :$$

a slice direction $I \in S^6$ with a normalized input coordinate u, the coordinate positioned at b, and the value produced there. Only the input is free. The positioned entry is determined by the position, and the value entry is the value $A_{\mathbb{O}}$ computes at the positioned entry: the second and third entries are one point of the equivariant realization's graph. The middle entry is a coordinate on its slice Riemann sphere; the third entry lies in $\mathbb{O}^*$, and this is where N enters the functor: at the pole the value is N itself, the value C1 assigns the pole. An arrow is a single element $\gamma \in G_2$ acting on the slice direction and carrying the rest of the triple with it. The disk action, conjugated by cayleyProjective, is the slice-preserving action on each Riemann sphere of the continuum in SphereWorld, the same complex map on every sphere: in components $D \cdot (I, u) = (I,\ D \cdot u)$, while γ moves the direction, $\gamma \cdot (I, u) = (\gamma I,\ u)$. Each acts on its own component, so the two commute:

$$D \cdot (\gamma \cdot q) = D \cdot (\gamma I,\ u) = (\gamma I,\ D \cdot u) = \gamma \cdot (I,\ D \cdot u) = \gamma \cdot (D \cdot q) :$$

the positioned entry moves coherently, and the value entry moves by equivariance (Definition 9.2(ii)). So $\mathcal{A}_A(b)$ is the action groupoid of G_2 on those triples. The fibres assemble the equivariant realization's graph, and their value entries are where condition C3 selects the residue-$\mathbb{C}$ zero states the concentricity theorem reads. In the Lean source the pairs are the states `AsectionState A`, whose point and value maps into $\mathcal{H}_1$ are `AsectionStateInput A` and `AsectionStateOutput A`, natural transformations checked with `AsectionState_input_then_equivariant`; the triples are `AsectionActionState`, their G_2 action groupoid is `AsectionActionStateFiber`, and each fibre stands at the object's own position: `AsectionActionFiber A X` is that groupoid at `projectiveObjectFrame A X`, which at N *is* D_A (`projectiveObjectFrame_north`).

Definition 9.19 (The transport)**.** On a base arrow $h : a \to b$, $\mathcal{A}_A(h)$ is a functor between the two fibre groupoids,

$$\mathcal{A}_A(h) : \mathcal{A}_A(a) \longrightarrow \mathcal{A}_A(b),$$

called *the transport*, and how it moves the states is read off A^{slice} at the same arrow. $A^{\text{slice}}(h)$ is the sphere-world arrow of h: its G_2 automorphism is the identity (`projectiveArrowHom_rot`), and its Möbius transformation is built from the factorization $h = o_b\, r_h\, o_a^{-1}$, each factor in place — o_a and o_b become the positions at a and at b, and r_h its image cayleyProjective(r_h) — equation (OS),

$$\big(\text{cayleyProjective}(o_b)\, D_A\big)\ \text{cayleyProjective}(r_h)\ \big(\text{cayleyProjective}(o_a)\, D_A\big)^{-1}.$$

All three factors are needed because the transformation has three jobs. A state at a stands in the position at a, and its transport must be a state at b, standing in the position at b: those are the outer two factors, and they are why both representatives appear. What h itself adds is its stabilizer factor r_h: that is the middle factor. Read right to left, the transformation undoes the position at a, acts by cayleyProjective(r_h), and positions the result at b.

A state carries two coordinates because the triple stores its graph point before and after the position: the input coordinate u is the normalized one, read on the one plane every slice shares, and the positioned coordinate is the position at a acting on it, the middle entry of the triple. The two meet the transformation differently. The positioned coordinate is $\mathrm{cayleyProjective}(o_a)\, D_A \cdot u$, so the transformation's rightmost factor, $\big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}$, multiplies the position it carries to the identity:

$$\Big(\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}\Big)$$
$$\cdot\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big) \cdot u \ =\ \big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h) \cdot u :$$

$\mathrm{cayleyProjective}(r_h)$ acts on u, and $\mathrm{cayleyProjective}(o_b)\, D_A$ places the result at b. The input coordinate stands before the positions, so the factor that reaches it is $\mathrm{cayleyProjective}(r_h)$ alone. And the value entry is computed at the transported positioned entry. On a triple at a the whole assignment reads

$$\Big(\,(I,u),\quad A^{\mathrm{slice}}(a)\cdot(I,u),\quad A_{\mathbb{O}}\big(A^{\mathrm{slice}}(a)\cdot(I,u)\big)\,\Big)$$
$$\longmapsto \Big(\,\big(I,\ \mathrm{cayleyProjective}(r_h)\cdot u\big),\quad A^{\mathrm{slice}}(b)\cdot\big(I,\mathrm{cayleyProjective}(r_h)\cdot u\big),$$
$$A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b)\cdot\big(I,\mathrm{cayleyProjective}(r_h)\cdot u\big)\big)\,\Big),$$

a state at b again; the total below shows each of the three entries forced.

Proposition 9.20 ($\mathcal{A}_A$ is a functor)**.** *The assignments $b \mapsto \mathcal{A}_A(b)$ and $h \mapsto \mathcal{A}_A(h)$ define a functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$, written `AsectionActionDiagramA` in the Lean source.*

Proof. On a fibre arrow the transport is the identity on γ, and the components show it. A fibre arrow moves only the direction, $\gamma\cdot(I,u) = (\gamma I, u)$; on the free input entry the transport moves only the coordinate u. Run the two on an input in both orders:

$$\gamma\cdot(I,u) = (\gamma I,\ u)\ \longmapsto\ \big(\gamma I,\ \mathrm{cayleyProjective}(r_h)\cdot u\big) = \gamma\cdot\big(I,\ \mathrm{cayleyProjective}(r_h)\cdot u\big) :$$

transporting the moved input lands on the move of the transported input, each element acting on its own component. The positioned and value entries follow by the commutation and equivariance displayed at the triple above, so the transport carries the fibre arrow γ to the same element γ between the transported states. The functor laws are the two laws of Lemma 9.15 carried through the homomorphism cayleyProjective, as at Proposition 9.17. Sweeping these fibres through the base by these transports completes the functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. □

$\mathcal{A}_A$ is the graph of A from the opening of the section, made categorical: input, position, value, the three columns of the picture, with the transport squares as the symmetries that carry graph points to graph points.

The total $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$

The Grothendieck construction gathers the indexed family built above — over each point b of the projective base the fibre $\mathcal{A}_A(b)$ — into a single category, and the construction asks four things of its input. First, the base is a groupoid. Second, the functor assigns a groupoid to each base object. Third, it assigns a functor between those groupoids to each base arrow. Fourth, the assignments are functorial, identities to identities and composites to composites. The first is already standing:

the base $\mathcal{B} = PGL(2,\mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$ is a groupoid because its arrows are group elements — an arrow is one projective element h with $h \cdot a = b$, composition is the group's multiplication, and the group inverse inverts every arrow. The remaining three are met by what the previous subsection built. A groupoid stands at each base object: the fibre $\mathcal{A}_A(b)$, the G_2 groupoid of the state triples, each triple consuming the two functors A^{slice} and $A_{\mathbb{O}}$ (Definition 9.18). A functor stands at each base arrow: the transport $\mathcal{A}_A(h) : \mathcal{A}_A(a) \to \mathcal{A}_A(b)$ (Definition 9.19). And the assignments are functorial (Proposition 9.20). With all four in hand, $\mathcal{T}_A$ is a well-defined total groupoid, and this subsection describes its objects and morphisms and proves the groupoid structure.

An object of the total $\int_{\mathcal{B}} \mathcal{A}_A$ is a pair (b, x): a point b of the real great circle and a state x of the fibre $\mathcal{A}_A(b)$ (Definition 9.18), the triple

$$x = \Big((I,u), \quad A^{\mathrm{slice}}(b) \cdot (I,u), \quad A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b) \cdot (I,u)\big) \Big)$$

positioned at b: one graph point of the A-section, placed. The pairing carries analytic content, and it is how this construction was first conceived: by GPV uniqueness, b carries the state's unique real fibre — every branch of the lift over it differs by one purely imaginary constant $2\pi i k$, so the whole fibre stands on one real number, the level the transport carries, and the exponential action every branch generates is one element (Theorem 6.5, Lemma 6.3).

The morphisms of the total are fixed by the transport (Definition 9.19), whose movers are read off A^{slice} at the same arrow: for two objects (a, x) and (b, x'),

$$\begin{aligned}\mathrm{Hom}_{\mathcal{T}_A}\big((a,x),(b,x')\big) = \big\{ (h,\gamma) \ &: h : a \to b \text{ in } \mathcal{B}, \\ &\gamma : \mathcal{A}_A(h)(x) \to x' \text{ in } \mathcal{A}_A(b) \big\}.\end{aligned}$$

The reason for the shape: the states x and x' — each one triple of the form of Definition 9.18 — stand in different groupoids, the fibres at a and at b, so no single group element joins them. The transport is the one passage between the fibres, and its value $\mathcal{A}_A(h)(x)$ — the moved triple, determined by h and x alone — is the only state at which a comparison with x' can start. So a morphism is two stages, base then fibre: h carries the state to the fibre at b, and γ compares directions there, the first entry (I, u) going to $(\gamma I, u)$ with the positioned and value entries moving by the commutation and equivariance displayed at $\mathcal{A}_A$.

Both components are concrete. The projective element h satisfies $h \cdot a = b$ and factors in exactly one way through the north stabilizer,

$$h = o_b \, r_h \, o_a^{-1}$$

(Lemma 9.15). The element $\gamma \in G_2$ satisfies $\gamma \cdot \mathcal{A}_A(h)(x) = x'$.

What remains to show is that these sets are the hom-sets of a category — the well-definedness of $\mathcal{T}_A$ — and it is two closures. The identity pair $(1_a, 1_x)$ must lie in $\mathrm{Hom}_{\mathcal{T}_A}\big((a,x),(a,x)\big)$: by Lemma 9.15, $r_{1_a} = 1$, so $\mathcal{A}_A(1_a)$ is the identity functor of the fibre and 1_x is a fibre arrow from $\mathcal{A}_A(1_a)(x) = x$ to x. And the composite of two pairs must again be one pair with one transport: $\mathcal{A}_A(h_1 \mathbin{⨟} h_2) = \mathcal{A}_A(h_2) \circ \mathcal{A}_A(h_1)$, the composition law of Lemma 9.15 carried through the transports. Proposition 9.22 below walks both closures on the states.

What the transport does to the triple is determined by two facts already on the page. The first is its definition on the positioned entry: the positioned entry moves by the Möbius transformation of $A^{\mathrm{slice}}(h)$, the equation (OS). The second is the form of a state: only the input is free, the positioned entry is the position acting on it, and the value entry is the value computed at the positioned entry — the linked triple of Definition 9.18. So the transported positioned entry is the Möbius transformation applied to the old one, and the question is which input coordinate now stands behind it. The position at b, $\mathrm{cayleyProjective}(o_b)\, D_A$, is invertible, so exactly one input coordinate does, and undoing the position computes it: undo the position at b, after the Möbius transformation, after the position at a — the three factors, one per line below. In the Möbius transformation the position at a meets its own inverse on the right, the inverse of the position at b meets the position at b on the left, and both products are the identity:

$$\begin{aligned}&\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)^{-1}\\ &\quad\Big(\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_h)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}\Big)\\ &\qquad\big(\mathrm{cayleyProjective}(o_a)\, D_A\big)\ =\ \mathrm{cayleyProjective}(r_h).\end{aligned}$$

What survives the two cancellations is $\mathrm{cayleyProjective}(r_h)$: the input coordinate behind the transported state is $\mathrm{cayleyProjective}(r_h)\cdot u$. With the input determined, the other two entries follow the linked form: the positioned entry is the position at b acting on that input — the same point the Möbius transformation produced, which is what the cancellation says — and the value entry is what $A_{\mathbb{O}}$ computes at that new positioned entry. Written out, the transported state is the triple

$$\begin{aligned}\mathcal{A}_A(h)(x) = \Big(\ &\big(I,\ \mathrm{cayleyProjective}(r_h)\cdot u\big),\quad A^{\mathrm{slice}}(b)\cdot\big(I, \mathrm{cayleyProjective}(r_h)\cdot u\big),\\ &A_{\mathbb{O}}\big(A^{\mathrm{slice}}(b)\cdot\big(I, \mathrm{cayleyProjective}(r_h)\cdot u\big)\big)\ \Big),\end{aligned}$$

standing in the fibre over b (`AsectionActionTransport_obj_input`, `AsectionActionTransport_obj_positioned`, `AsectionActionTransport_obj_value`; this cancellation at every base arrow is `orbitStabilizerActionSquare` with its identity `projectiveArrowElement_frame_compat`, and `ActionTransportSquare.apply` reads it at every coordinate).

Two uniquenesses stand behind every morphism, both anchored at N. By GPV uniqueness, what the A-section consumes over each real base object is unique: the tame continuous lift Γ_A, determined by its initial value in the exponential fibre and continued to N, where every run arrives at D_A with the real level $\operatorname{Re}\lambda_A$ (the GPV transport above). By stabilizer uniqueness (Lemma 9.15), the projective element h factors in exactly one way through the north stabilizer, $h = o_b\, r_h\, o_a^{-1}$: one factorization, one stabilizer factor. So the data a morphism consumes are forced, the analytic datum by GPV and the group datum by the fixed representatives, and the morphism itself is just the pair (h, γ).

Remark 9.21 (N in the total). N itself is never consumed by a disk action in the projective base: the runs end at the pole, and the section's value at the domain point N is the marked datum (Remark 1.7). What stands at N is the unique GPV real fibre: any two lifts arriving there differ by one purely imaginary constant $2\pi i k$, so the whole fibre stands on the one real number $\operatorname{Re}\lambda_A$ (Theorem 6.5, Proposition 6.6; [21, Cor. 4.13]; `joint_winding`, `joint_re_eq`).

Proposition 9.22 ($\mathcal{T}_A$ is a groupoid). *$\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ is a groupoid: composition is group multiplication in* $\mathrm{PGL}(2,\mathbb{R})$ *and in* G_2*,*

$$(h_1, \gamma) \mathbin{⨟} (h_2, \gamma') = \big(h_1 \mathbin{⨟} h_2,\ \gamma'\gamma\big),$$

well defined because the transports compose in the slice Möbius group, the classes of $\mathrm{GL}(2,\mathbb{C})$*:* cayleyProjective *carries the composite's stabilizer factor $r_{h_1 ⨟ h_2} = r_{h_2} r_{h_1}$ (Lemma 9.15) onto the composed Möbius transformations. Every morphism has the inverse $(h,\gamma)^{-1} = (h^{-1}, \gamma^{-1})$.*

Proof. Identities first. At the identity arrow 1_a both ends are a and the stabilizer factor is $r_{1_a} = 1$ (Lemma 9.15), so both movers of the transport collapse: the input entry moves by $\mathrm{cayleyProjective}(1) = 1$, and the Möbius transformation is equation (OS) at $r_{1_a} = 1$, which cancels,

$$\big(\mathrm{cayleyProjective}(o_a)\, D_A\big)\ \mathrm{cayleyProjective}(1)\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1} = 1.$$

Both movers are the identity, so every entry stands where it stood and the value entry is recomputed at that same positioned entry: $\mathcal{A}_A(1_a)$ is the identity functor of the fibre at a and $(1_a, 1_x)$ is the identity morphism of (a, x) — the first closure promised at the hom-set (`AsectionActionTransport_id`). Next a fact the composition needs: the transport $\mathcal{A}_A(h)$ is the identity on fibre arrows. A fibre arrow is one element $\gamma \in G_2$ acting on the direction; the transport moves coordinates, the input entry by cayleyProjective(r_h) and the positioned entry by the Möbius transformation; each acts on its own component, the computation displayed at the triple, so

$$\mathcal{A}_A(h)\big(\gamma : x \to y\big) = \gamma : \mathcal{A}_A(h)(x) \to \mathcal{A}_A(h)(y),$$

the same group element between the transported states (`AsectionActionTransport`). Composition is the same two stages run for a composable pair, $(h_1, \gamma) : (a, x) \to (b, x')$ followed by $(h_2, \gamma') : (b, x') \to (c, x'')$. A composite must again be one member of the hom-set displayed above: one projective element with one G_2 element in the target fibre. The projective elements compose to $h_1 \mathbin{⨾} h_2$, whose stabilizer factor is $r_{h_2} r_{h_1}$ by Lemma 9.15: the composite projective element is $o_c\,(r_{h_2} r_{h_1})\, o_a^{-1}$. For the G_2 element, γ lives in the fibre over b, the transport $\mathcal{A}_A(h_2)$ carries it into the fibre over c as the same group element, and γ' follows it:

$$\mathcal{A}_A(h_1 \mathbin{⨾} h_2)(x) = \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x)\big) \xrightarrow{\gamma} \mathcal{A}_A(h_2)(x') \xrightarrow{\gamma'} x'',$$

the equality being $\mathcal{A}_A(h_1 ⨾ h_2) = \mathcal{A}_A(h_2) \circ \mathcal{A}_A(h_1)$: both sides move every entry by cayleyProjective of the same stabilizer factor, since $r_{h_1 ⨾ h_2} = r_{h_2} r_{h_1}$ by Lemma 9.15 — the second closure promised at the hom-set (`AsectionActionTransport_comp`). So composition is group multiplication in $\mathrm{PGL}(2, \mathbb{R})$ and in G_2, and the inverse inverts both:

$$(h_1, \gamma) \mathbin{⨾} (h_2, \gamma') = \big(h_1 \mathbin{⨾} h_2,\ \gamma'\gamma\big), \qquad (h, \gamma)^{-1} = \big(h^{-1},\ \gamma^{-1}\big),$$

and the inverse (h^{-1}, γ^{-1}) composes with (h, γ) to the identity pair on both sides, shown by the two laws. The base is a groupoid, so the two composites of $h : a \to b$ with its inverse are the two identity arrows, $h \mathbin{⨾} h^{-1} = 1_a$ and $h^{-1} \mathbin{⨾} h = 1_b$. Take stabilizer factors:

$$r_{h^{-1}}\, r_h = r_{h ⨾ h^{-1}} = r_{1_a} = 1, \qquad r_h\, r_{h^{-1}} = r_{h^{-1} ⨾ h} = r_{1_b} = 1.$$

In each chain the first equality is the composition law, the second is the composite being the identity arrow, and the third is the identity law — the two laws of Lemma 9.15. Together the chains say $r_{h^{-1}} = r_h^{-1}$: the stabilizer factor of the inverse arrow is the inverse stabilizer factor. The two transports in question are $\mathcal{A}_A(h)$ and $\mathcal{A}_A(h^{-1})$, the functors between the fibre groupoids $\mathcal{A}_A(a)$ and $\mathcal{A}_A(b)$, and each moves a state's entries by cayleyProjective of its stabilizer factor — the transported triple above. The homomorphism cayleyProjective carries each equation onto those images, so each composite transport moves every entry by the image of 1, which moves nothing: $\mathcal{A}_A(h \mathbin{⨾} h^{-1})$ and $\mathcal{A}_A(h^{-1} ⨾ h)$ are the identity functors of the two fibres. This says what every transport is in **Grpd**: an isomorphism — a functor with a two-sided inverse functor, $\mathcal{A}_A(h^{-1})$ — between the fibre groupoids $\mathcal{A}_A(a)$ and $\mathcal{A}_A(b)$, an automorphism of the fibre $\mathcal{A}_A(a)$ when $b = a$. The reason is the base: it is a groupoid, every h has h^{-1}, and the functor laws turn the base inverse into the inverse functor. So the total is a groupoid: its composition, identities, and inverses are the pair formulas of the statement. In the Lean source the construction is Mathlib's `CategoryTheory.Grothendieck` applied to `AsectionActionDiagram A`, and the groupoid classification is `grothendieckGrpdGroupoid`. □

Every fibre stands at its position, D_A along its fixed representative, and at N the position *is* D_A; every transport conjugates the stabilizer factor of a base arrow between two such positions; every value entry is the equivariant realization computed at a positioned point. The total is the A-section's graph made one category: what moves from fibre to fibre is D_A's data, and the populated residue-$\mathbb{C}$ zero-sphere states are the degenerate-fibre outputs of this completed construction, produced by C3–C4. The total is read in two ways, and both are functors it comes with. The projection to the base, $(b, x) \mapsto b$ and $(h, \gamma) \mapsto h$, forgets the state (`CategoryTheory.Grothendieck.forget`); over it the total holds one fibre per point of the real great circle. And the components diagram $\pi_0 \circ \mathcal{A}_A : \mathcal{B} \to \mathbf{Set}$ sends each base object to the set of components of its fibre; its colimit is what the readout consumes (Lemma 10.3) — π_0 of the total is the colimit of the fibrewise components, which is how the next section turns connectedness into one real number.

Definition 9.23 (The projective base, the A-section functor, and the total category)**.** The base is the projective action groupoid $\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$. The A-section functor is the assignment constructed in this section — on objects the G_2 action groupoid of state triples positioned at each base object, on arrows transport by the Möbius transformation (OS) — written `AsectionActionDiagram A`, $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. The total value-transport category is $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$.

10 The concentricity theorem

We now have the worlds and the base, so this section defines the class of functions precisely and proves the theorem. Two facts about slice-preserving functions come first: the Weierstrass factorization that gives condition C3 its meaning, and the fact that a residue-$\mathbb{C}$ zero is a whole 6-sphere, a single connected component of the octonionic world. Then the categorical engine, Riehl's computation of the components of a Grothendieck construction as a colimit, which sorts the assembled total into the classes linked by the section's real-value transports. With these in hand we define an A-section by its four conditions C1–C4, state and prove the Concentricity Theorem, and record the corollary that carries its conclusion back into the language of classical zeros.

Proposition 10.1 (Slice-regular Weierstrass factorization; the content of C3)**.** *Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as*

$$A = q^m \, R \, S \, h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order-m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A, S is a factor vanishing exactly at its spherical (residue-$\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds including the case where either family of zeros is infinite*.*

Proof. The proposition is general: A here is any member of $\mathcal{R}$, the hypotheses C1–C4 play no role in the argument, and either family of zeros may be finite, infinite, or empty. Every analytic step happens one algebra down, at the stem; the slice formalism then lifts the finished factorization to $\mathbb{O}^*$.

The reduction to the stem. By the I-independent lift (Definition 6.4), A is induced by one holomorphic stem $F = F_1 + \iota F_2$ with F_1, F_2 $\mathbb{R}$-valued:

$$A(x + Iy) = F_1(x + iy) + I \, F_2(x + iy), \qquad \text{the same components for every } I.$$

Real components give $F(\bar z) = \overline{F(z)}$, so the zero set of F is conjugation-symmetric: the real zeros ρ_k stand individually, and the non-real zeros stand in conjugate pairs $\{\rho_j, \bar\rho_j\}$. Under the lift a real zero is a residue-$\mathbb{R}$ zero of A, and one conjugate pair is one zero-sphere, its whole G_2-orbit (Lemma 10.2).

The classical factorization, symmetrized. A holomorphic function g has real Taylor coefficients exactly when it satisfies the conjugation symmetry $\overline{g(\bar z)} = g(z)$; this is the form in which realness is tracked below, and it is the symmetry the stem already has. The classical Weierstrass theorem factors F over its zero set: with the primary factors

$$E_p(w) = (1-w)\exp\Big(w + \tfrac{w^2}{2} + \cdots + \tfrac{w^p}{p}\Big)$$

and the integers p_n chosen so that $\sum_n (r/|\rho_n|)^{p_n+1}$ converges for every $r > 0$, the product converges normally on compact sets and

$$F(z) = z^m \prod_k E_{p_k}(z/\rho_k) \prod_j \Big(E_{p_j}(z/\rho_j)\, E_{p_j}(z/\bar\rho_j)\Big)\, h_0(z),$$

with h_0 holomorphic and zero-free, the two factors of each conjugate pair multiplied as displayed. Each primary factor is built from real coefficients, so conjugating it conjugates its argument: $E_p(\bar w) = \overline{E_p(w)}$. For a real zero ρ_k the factor $E_{p_k}(z/\rho_k)$ inherits the symmetry in z directly. For a conjugate pair, write $P_j(z) = E_{p_j}(z/\rho_j)\, E_{p_j}(z/\bar\rho_j)$ and compute, the two factors exchanging under conjugation:

$$\overline{P_j(\bar z)} = \overline{E_{p_j}(\bar z/\rho_j)}\;\overline{E_{p_j}(\bar z/\bar\rho_j)} = E_{p_j}(z/\bar\rho_j)\, E_{p_j}(z/\rho_j) = P_j(z),$$

real coefficients again, visible at the degree-two core:

$$\big(1 - z/\rho\big)\big(1 - z/\bar\rho\big) = 1 - 2\,\mathrm{Re}(1/\rho)\, z + |\rho|^{-2} z^2.$$

Finally, F and every factor satisfy the symmetry, and the symmetry passes to quotients, so the zero-free h_0 satisfies it as well. Every factor of the factorization, h_0 included, is therefore a holomorphic stem with real components.

The lift. The stem formalism of Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2] is stated for real alternative $*$-algebras, hence for $\mathbb{O}$, and for *arbitrary* stems: the formalism itself never requires real components. Real components are what the previous step produced, and they buy exactly two things: the induced section is intrinsic, and on slice-preserving sections the regular product is the pointwise product (Theorem 5.2). So the factorization lifts factor by factor to $A = q^m RSh$ on $\mathbb{O}^* \setminus \{N\}$: R the lift of the real-zero product, S the lift of the paired products, each vanishing exactly on its zero-sphere, and h the lift of h_0, never vanishing. The compactified frame in which this is read, the slice spheres of $\mathbb{O}^*$ with $\varphi_v(\infty) = N$, is GPV's (Remark 3.6; [17, Prop. 4.1, Thm. 4.2]): slice preservation itself carries their facts, before and independently of the transport sections. Convergence transfers because the components are direction-blind: for every I,

$$|A(x+Iy)|^2 = F_1(x+iy)^2 + F_2(x+iy)^2 = |F(x+iy)|^2,$$

so a normal bound for the stem products is the same bound on every slice at once. The lift is the stem formalism of Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2], stated for real alternative $*$-algebras and hence for $\mathbb{O}$.

Placement in the literature, and the instance consumed. Over $\mathbb{H}$ the factorization is the theorem of Gentili–Vignozzi [10], with the slice-preserving case in Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2]. Over $\mathbb{O}$ we state and prove it by the reduction above: for an intrinsic section the analysis is the classical theorem at the stem, and the octonionic content is carried entirely by the lift. The instance the paper consumes is the A-section's: there the pair R, S is the residue split named in C3, R carrying the residue-$\mathbb{R}$ (classically trivial) zeros and S the residue-$\mathbb{C}$ (classically nontrivial) zero-spheres, the latter infinite in number by C4. ☐

Lemma 10.2 (Residue-$\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$)**.** *For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of the translation groupoid $\mathcal{H}_1$.*

Proof. A is slice-preserving, so on a sphere $\sigma + \gamma S^6$ ($\gamma > 0$) it has the form $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ with α, β real (the real stem, [16]). Suppose A vanishes at one point of the sphere: $\alpha + v\beta = 0$. The real part kills the first component: v is imaginary, so $\mathrm{Re}(\alpha + v\beta) = \alpha = 0$. The modulus of what remains kills the second: $|v\beta| = |v|\,|\beta| = |\beta|$, so $\beta = 0$. Neither component depends on v: the zero occurs at every point of $\sigma + \gamma S^6$ or at none (the spherical-zero structure, [1]); real zeros are isolated real points (residue field $\mathbb{R}$).

The sphere $\sigma + \gamma S^6$ lies in $\mathbb{O}^*$ off the real axis, and it is one G_2-orbit: G_2 acts transitively on S^6 (Theorem 7.5, stabiliser SU(3)), fixing σ and γ and sweeping the direction. The connected components of $\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*$ are its orbits, the translation-groupoid device $\pi_0(G \ltimes X) = X/G$ (Quillen [22, §1]), so each zero-sphere is one component of $\mathcal{H}_1$. Therefore the residue-$\mathbb{C}$ zeros, the non-real ones, are exactly the 6-sphere components of $A^{-1}(0)$. ☐

Lemma 10.3 (π_0 of a Grothendieck construction)**.** *For a functor $F : \mathcal{B} \to$ Grpd, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\Big(\int_{\mathcal{B}} F\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}}\,(\pi_0 \circ F).$$

Proof. By Riehl's proof of Lemma 8.3.4, for any $X : \mathcal{C} \to \mathbf{Set}$ each arrow of the category of elements imposes exactly the condition that the corresponding elements be identified in every cone, so that $\pi_0(\mathrm{el}\,X) \cong \mathrm{colim}_{\mathcal{C}}\,X$ [23, p. 102]. Taking $X = \pi_0 \circ F$, the colimit performs exactly the identifications generated by the maps of F. In the Lean development this equivalence is `pi0GrothendieckEquiv`, built from the canonical cocone `pi0Cocone` and `colimit.desc` over Mathlib's `colimit_sound` and `colimit_eq`. ☐

Definition 10.4 (A-sections)**.** An *A-section* is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) with the following four properties.

(C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple, lies at a real point, and its value there is the point at infinity $\infty = N$. Meromorphy is a finite-point condition; at the domain point N itself no regularity — holomorphy, meromorphy, or continuity — is claimed (Remark 1.7).

(C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, A exponentiates one prime-indexed degenerate logarithmic base — a family *made of* the primes, not merely indexed by them: $A = \exp\big(\sum_p \ell_p\big)$, the sum over the infinite set of primes, where each ℓ_p is slice-preserving, slice-regular, zero-free on Ω_0, and of *prime-power logarithmic form*

$$\ell_p(q) = \sum_{k \geq 1} \frac{a_{p,k}}{k}\, p^{-kq},$$

with real coefficients $a_{p,k}$ bounded uniformly in p and k, the family summable *locally normally* on Ω_0 (the sense in which the cited Euler products converge; Theorem 1.2). In particular A is zero-free on Ω_0; every term decays as $\operatorname{Re} q \to +\infty$, so $A \to 1$ there and the marked value at N is 1 (Remark 1.7); and the logarithm of A is *one* exponential base — prime-power frequencies, constant-scale coefficients, no translated anchor, no additive pole term, the pole of C1 arising instead from the divergence of the prime sum at the boundary of the half-space, as for ζ at $s = 1$ — the single base the unique tame continuous lift carries on the degenerate set (Theorems 6.2, 6.5). The prime indexes its term through its own powers, not as a label: each ℓ_p is a power series in p^{-q} for its own prime p, so a family merely enumerated by primes, whose terms are not such power series, is not of this form.

(C3) *Infinite Weierstrass factorization.* Over the *full divisor* of A, including the single pole of (C1), at the real point p_0, multiplicity -1, carried by the explicit left factor, on $\mathbb{O}^* \setminus \{p_0\}$,

$$(q - p_0)\, A \;=\; q^m\, R\, e^g \prod_n \mathcal{E}(\,\cdot\,; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the *nonzero* residue-$\mathbb{R}$ (real) zeros of A, vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue-$\mathbb{C}$ zero-spheres of A, and $\mathcal{E}(\,\cdot\,; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite, the product converging *locally normally* on $\mathbb{O}^* \setminus \{p_0\}$ (the convergence of Proposition 10.1). (The left factor makes "full divisor" literal: the divisor of a meromorphic section includes its single pole; classically, Hadamard factors $(s-1)\zeta(s)$.)

(C4) *Infinitely many residue-$\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$; [2]); residue-$\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Definition 10.5 (The residue subdiagram and its inclusion)**.** The *residue locus* of an A-section is $\{z : A(z) = 0,\ \operatorname{Im} z > 0\}$ (`CResidueZeroLocus`). It is read at the one north object, and everything already built says why that one object suffices. The compactified slices share the single point N (Remark 3.6; [17, Prop. 4.1, Thm. 4.2]); the fibre of the total over N stands at D_A itself; and condition C3 presents the arriving multiplier there in Weierstrass form, its sphere factors enumerating exactly these zeros (Proposition 10.1). So the locus is selected once, at N, where C3 presents D_A in Weierstrass form.

The *residue subdiagram* $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ (`AsectionCResidueDiagram`) is the preimage of the states this locus selects at N — a preimage taken in the total action groupoid, not of a set: a state belongs exactly when a base arrow carries a selected north state onto it.

It has the same source as $\mathcal{A}_A$, the projective action groupoid $\mathcal{B}$, and its values are again action groupoids of G_2 — the selection cuts down which states are present without altering what acts on them.

On a base object a, the value $\mathcal{R}_A(a)$ is the full subgroupoid of $\mathcal{A}_A(a)$ on the reached states: a state x of $\mathcal{A}_A(a)$ belongs when there are a north residue state x_N and a base arrow $h_1 : N \to a$ — one projective transformation carrying the north point to a — with $\mathcal{A}_A(h_1)(x_N) = x$. Membership is the whole selection: nothing further is added to the object. An arrow is an arrow of $\mathcal{A}_A(a)$ between two such states, so again a single element $\gamma \in G_2$: the selection is on objects, and G_2 still acts.

On a base arrow $h_2 : a \to b$, the value $\mathcal{R}_A(h_2)$ is the restriction of the ambient transport $\mathcal{A}_A(h_2)$, which is transport by the Möbius transformation (OS) between the positions at a and b. That restriction lands in the subgroupoid because a reached state stays reached, from the *same* north state along the composite $h_1 \mathbin{⨾} h_2 : N \to b$. Functoriality of $\mathcal{A}_A$ and then the reaching equation give

$$\mathcal{A}_A(h_1 \mathbin{⨾} h_2)(x_N) = \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x_N)\big) = \mathcal{A}_A(h_2)(x).$$

So the north state is what is preserved, and the reaching projective transformation is what accumulates. Its inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$ (`AsectionCResidueInclusion`) sends each reached state to itself as an ambient state and is the identity on arrows; it is fibrewise fully faithful onto that inverse image, and its naturality squares retain the ambient A-section transport itself.

Lemma 10.6 (The C-residue inclusion)**.** *For an A-section A, the total inclusion*

$$\textstyle\int_{\mathcal{B}} \mathcal{R}_A \xhookrightarrow{\int \iota_A} \int_{\mathcal{B}} \mathcal{A}_A = \mathcal{T}_A$$

is full and faithful: an isomorphism onto its image, which consists exactly of the selected residue states together with the restricted A-section transports between them.

Proof. Two things are to be shown, and each is read off a structure already built. Fibrewise: the inclusion of each fibre is full and faithful, which is the statement that the selection touched objects only and no hom-set was cut. Over the base: the fibrewise inclusions commute with every A-section transport, so they assemble into the total inclusion. The inputs are three: an arrow of an action groupoid is one group element with its transport equation; the fibres of $\mathcal{R}_A$ are full subgroupoids (Definition 10.5); and the transports obey the two laws of Lemma 9.15.

The fibrewise inverse image. For each base object a, an object of $\mathcal{R}_A(a)$ is an ambient state $x \in \mathcal{A}_A(a)$ reached from the north residue locus: there are a north residue state x_N and a projective element $h_1 : N \to a$, one element of $\mathrm{PGL}(2, \mathbb{R})$ with $h_1 \cdot N = a$, satisfying

$$\mathcal{A}_A(h_1)(x_N) = x. \tag{W}$$

This is the object property `IsCResidueState A X`. The reaching arrows are elements of the one projective base $\mathcal{B}$, over whose objects stands the unique prime-power GPV lift the A-specific action exponentiates: the preimage is of the action on that lift, and the base is already fixed by it. The fibre $\mathcal{R}_A(a)$ is the full subgroupoid of $\mathcal{A}_A(a)$ on precisely these objects, and an arrow of $\mathcal{R}_A(a)$ is one element $\gamma \in G_2$: the group action already built. Its inclusion

$$\iota_{A,a} : \mathcal{R}_A(a) \hookrightarrow \mathcal{A}_A(a)$$

sends each reached state to itself as an ambient state. For two such objects x, y, the induced map

$$\mathrm{Hom}_{\mathcal{R}_A(a)}(x, y) \longrightarrow \mathrm{Hom}_{\mathcal{A}_A(a)}(\iota_{A,a}x, \iota_{A,a}y) \tag{H}$$

is a bijection, and the reason is definitional. An arrow of $\mathcal{A}_A(a)$ between the two states is one element $\gamma \in G_2$ with $\gamma \cdot x = y$; that set is what the hom-set is. The fibre $\mathcal{R}_A(a)$ is a *full* subgroupoid: the selection removed objects and declared the hom-set between any two kept states to be the ambient one, unchanged. So the two sides of (H) are the same set,

$$\mathrm{Hom}_{\mathcal{R}_A(a)}(x, y) = \{\gamma \in G_2 : \gamma \cdot x = y\} = \mathrm{Hom}_{\mathcal{A}_A(a)}(\iota_{A,a}x, \iota_{A,a}y),$$

and (H) is the identity assignment $\gamma \mapsto \gamma$ on it. Surjectivity of (H) is fullness of $\iota_{A,a}$; injectivity is faithfulness.

Transport of the inverse image. Let $h_2 : a \to b$ be a projective element of the base, one element of $\mathrm{PGL}(2, \mathbb{R})$ with $h_2 \cdot a = b$. The ambient transport it induces is the Möbius transformation (OS), the stabilizer factor conjugated between the two positions:

$$\big(\mathrm{cayleyProjective}(o_b)\, D_A\big)\ \mathrm{cayleyProjective}(r_{h_2})\ \big(\mathrm{cayleyProjective}(o_a)\, D_A\big)^{-1}.$$

A reached state stays reached: its reaching arrow composes with the base arrow, and the composite $h_1 \fatsemi h_2 : N \to b$ is one projective element again, the matrix product $h_2 h_1$. Drawn over the base, with the composite as the bent arrow and the dashed strokes tying each state to the base object whose fibre holds it:

$$\begin{tikzcd}
x_N \arrow[r, mapsto, "\mathcal{A}_A(h_1)"] \arrow[d, dash, dashed] & x \arrow[r, mapsto, "\mathcal{A}_A(h_2)"] \arrow[d, dash, dashed] & \mathcal{A}_A(h_2)(x) \arrow[d, dash, dashed] \\
N \arrow[r, "h_1"'] \arrow[rr, bend right, "h_1 \fatsemi h_2"'] & X \arrow[r, "h_2"'] & Y
\end{tikzcd}$$

The first passage lands on x by the reaching equation (W) itself. So if x is reached as in (W), then $\mathcal{A}_A(h_2)(x)$ is reached from the same north state x_N along the composite $h_1 \fatsemi h_2 : N \to b$: writing the two equalities with their reasons,

$$\begin{aligned}
\mathcal{A}_A(h_1 \fatsemi h_2)(x_N) &= \mathcal{A}_A(h_2)\big(\mathcal{A}_A(h_1)(x_N)\big) && \text{the functor law } \mathcal{A}_A(h_1 \fatsemi h_2) = \mathcal{A}_A(h_2) \circ \mathcal{A}_A(h_1), \\
&= \mathcal{A}_A(h_2)(x) && \text{the reaching equation (W).}
\end{aligned} \tag{C}$$

This is the calculation recorded by `cResidue_lands`. It defines the restricted transport

$$\mathcal{R}_A(h_2) : \mathcal{R}_A(a) \longrightarrow \mathcal{R}_A(b) :$$

on objects its membership is (C)'s composite reaching, and on arrows it applies the ambient transport $\mathcal{A}_A(h_2)$. Composition in the projective base therefore supplies closure of the inverse-image system under every A-section transport.

The natural inclusion. The fibrewise inclusions assemble over $\mathcal{B}$ in the square

$$\begin{tikzcd}
\mathcal{R}_A(a) \arrow[r, "\mathcal{R}_A(h_2)"] \arrow[d, hook, "\iota_{A,a}"'] & \mathcal{R}_A(b) \arrow[d, hook, "\iota_{A,b}"] \\
\mathcal{A}_A(a) \arrow[r, "\mathcal{A}_A(h_2)"'] & \mathcal{A}_A(b).
\end{tikzcd} \tag{Nat}$$

For an object x, both routes in (Nat) have underlying state $\mathcal{A}_A(h_2)(x)$; for an arrow α, both routes have underlying arrow $\mathcal{A}_A(h_2)(\alpha)$. The upper route additionally lands in the subgroupoid, by the composite reaching established in (C). Hence

$$\mathcal{R}_A(h_2) \fatsemi \iota_{A,b} = \iota_{A,a} \fatsemi \mathcal{A}_A(h_2). \tag{Inc}$$

and these equations for every h_2 are the naturality of $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$.

The total inclusion. The preimage assembles over the base into a total of its own. Its objects are the positioned squares of the C-residue image, and a morphism is a pair (h, γ): one element of $\mathrm{PGL}(2, \mathbb{R})$ and one element of G_2, the value transports of $\mathcal{T}_A$ between them. The total inclusion (`AsectionCResidueInclusionTotal` in the Lean source, the Grothendieck functoriality of ι_A applied to it) sends (a, x) to $(a, \iota_{A,a} x)$ and leaves the point a unchanged; on a Grothendieck morphism it leaves the base element unchanged and applies the fibre inclusion only to the compatible transport. Consequently equality after inclusion forces equality of the base elements and, by fibrewise faithfulness, equality of the transport elements: the total inclusion is faithful. Conversely, an ambient total morphism between two objects in the image is already such a pair (h, γ), one projective element and one G_2 element between selected target states; fibrewise fullness gives its preimage in $\mathcal{R}_A$, with the same h. Hence the total inclusion is full (`AsectionCResidueInclusionTotal_full` and `AsectionCResidueInclusionTotal_faithful`). It is therefore an isomorphism onto its image, and the C-residue total is that image itself: the selected objects together with the restricted A-section transports between them. □

Lemma 10.7 (Transitivity of the C-residue total)**.** *For an A-section A, the total inverse-image groupoid*

$$\int_{\mathcal{B}} \mathcal{R}_A$$

is transitive: any two of its objects are joined by a morphism.

Proof. The proof is two stages: every object of $\int_{\mathcal{B}} \mathcal{R}_A$ is joined to a north residue state, and any two north residue states are joined in the fibre at N; composing the stages joins any two objects.

Every object is joined to its north state. Let $P \in \int_{\mathcal{B}} \mathcal{R}_A$. By Definition 10.5 its state is reached: there are a north residue state x_N and a base arrow $h_1 : N \to P.\text{base}$ with

$$\mathcal{A}_A(h_1)(x_N) = P.\text{fiber}.$$

The transport $\mathcal{A}_A(h_1)$ is an isomorphism of fibres (Proposition 9.22), so the same equality reads backwards, $\mathcal{A}_A(h_1^{-1})(P.\text{fiber}) = x_N$, and the pair $(h_1^{-1}, 1_{x_N})$ is a morphism $P \to (N, x_N)$ of the total. Both ends are residue states, and the inclusion is full and faithful (Lemma 10.6), so this morphism lies in $\int_{\mathcal{B}} \mathcal{R}_A$. What remains is the second stage: joining two north residue states in the fibre at N.

The state is a triple. At N the A-section action functor records

$$(I, u) \longmapsto \Big(\underbrace{(I, u)}_{\text{input}},\ \underbrace{D_A \cdot (I, u)}_{\text{positioned}},\ \underbrace{A_{\mathbb{O}}(D_A \cdot (I, u))}_{\text{realized value}} \Big). \tag{A}$$

The three entries carry the three things the comparison needs, and they carry them separately. The residue condition of $\mathcal{R}_A(N)$ selects on the *middle* entry: a north residue state is one whose positioned coordinate is an authored C3 coordinate. That is the entry the inverse-image construction is built on.

The two states, and where they differ. Write the two supplied states as evaluations of (A):

$$\begin{aligned} x_N &= \big((I_1, u_1),\ z_1,\ A_{\mathbb{O}}(z_1)\big), \quad z_1 = D_A \cdot u_1, \\ y_N &= \big((I_2, u_2),\ z_2,\ A_{\mathbb{O}}(z_2)\big), \quad z_2 = D_A \cdot u_2, \end{aligned} \tag{P}$$

with z_1, z_2 the authored C3 coordinates their residue condition selects and $I_1, I_2 \in S^6$ their slice directions. Since D_A is invertible, each middle entry has exactly one input behind it, so (P) determines u_1 and u_2 rather than choosing them. No equality between these two inputs is assumed: their positioned coordinates may differ, while their realized values satisfy the common residue-zero condition.

The inputs u_1 and u_2 are arbitrary and generally different. What the two states share is the north object N, the position D_A there, and the construction being evaluated — not a fibre input over the source object 0.

Euler at 0, Weierstrass at the one north object. For each state separately, condition C2 supplies the arithmetic Euler presentation

$$\Gamma_A(t) = \sum_p \ell_p(\delta(t)), \qquad e^{\Gamma_A(t)} = A(\delta(t)),$$

with the prime index retained inside the complete summation. Local normal convergence makes Γ_A continuous, the zero-free half-space keeps its value in $\mathbb{C}^\times$, and GPV uniqueness fixes the lift once its initial value is fixed. Condition C1 continues that Euler presentation to N — the continuation happens at the finite pole, and N receives its output by positioning (Remark 1.7) — where C3 supplies its Weierstrass presentation. Thus the Euler product is present at 0 and, for each of the two arbitrary residue inputs, the same continued action is presented in Weierstrass form at N. The two evaluations share N and D_A while retaining their own semantic inputs u_1 and u_2.

The comparison at the one north object. There is one north object, $N = \infty$, with its own self-map. Each of the two states is one evaluation of the same construction, and its run arrives at N with the same element and the same real value:

$$D(\Gamma_1(1)) = D_A = D(\Gamma_2(1)), \qquad \operatorname{Re}\Gamma_1(1) = \operatorname{Re}\lambda_A = \operatorname{Re}\Gamma_2(1),$$

while the two runs differ at N by a whole winding,

$$\Gamma_2(1) - \Gamma_1(1) = 2\pi i k, \qquad k \in \mathbb{Z}.$$

At the one north object the two evaluations differ only by the whole winding, and the exponential action does not see it. The coordinate each run carries to N is its own state's positioned C3 coordinate, read at D_A by (OS). The two arbitrary states are connected by the action's own morphisms: the north self-map k, read through D_A, carries the first positioned coordinate to the second,

$$\left(D_A \operatorname{cayleyProjective}(k)\, D_A^{-1}\right) \cdot z_1 = z_2,$$

the comparison square at the one north object, its verticals D_A:

$$\begin{tikzcd}
u_1 \arrow[r, "\operatorname{cayleyProjective}(k)"] \arrow[d, "D_A"'] & u_2 \arrow[d, "D_A"] \\
z_1 \arrow[r, "D_A \operatorname{cayleyProjective}(k)\, D_A^{-1}"'] & z_2
\end{tikzcd}$$

This is the coordinate component of the comparison at N.

The direction, and why it is available here. The comparison square moves the positioned coordinate. The direction entry remains, and it is the entry the functor swept.

What $\mathcal{A}_A$ sweeps is the slice, and what indexes the slices is the direction: I ranges over S^6, the unit imaginary octonions, D_A positions the coordinate in the slice that I selects, and $A_{\mathbb{O}}$ realizes the value there. So the directions carried by x_N and y_N are not labels attached afterwards — they are the indices the functor swept over, and every one of them is a unit imaginary octonion because that is what a slice direction is (Definition 9.1). This is exactly the set on which G_2 is transitive (Theorem 7.5), so it supplies $\gamma \in G_2$ with

$$\gamma I_1 = I_2.$$

The ambient total cannot take this step. Its objects are all the states of $\mathcal{A}_A$, and G_2 moves a state only within the sphere its direction already lies on. The inverse image, by contrast, is taken over precisely what the sweep produces — the residue states on the C3 zero spheres — so the transitivity of G_2 on the unit imaginary octonions acts on the whole of what $\int_{\mathcal{B}} \mathcal{R}_A$ contains, and on nothing larger. That is why this comparison is available there and not in $\int_{\mathcal{B}} \mathcal{A}_A$.

The direction is γ's own action, and the transported coordinate equality completes $\gamma \cdot \mathcal{A}_A(k)(x_N) = y_N$: γ moves the direction and keeps the coordinate, so it is the fibre arrow

$$\gamma : \mathcal{A}_A(k)(x_N) \longrightarrow y_N. \tag{γ}$$

Composing the stages. The two stages together join arbitrary P and Q. Stage one gives $(h_1^{-1}, 1_{x_N}) : P \to (N, x_N)$, and at Q the reached equality $\mathcal{A}_A(h_2)(y_N) = Q.\mathrm{fiber}$ makes $(h_2, 1_{Q.\mathrm{fiber}})$ a morphism $(N, y_N) \to Q$. Stage two gives $(k, \gamma) : (N, x_N) \to (N, y_N)$, the fibre arrow being (γ). Their composite

$$P \xrightarrow{(h_1^{-1},\, 1_{x_N})} (N, x_N) \xrightarrow{(k,\gamma)} (N, y_N) \xrightarrow{(h_2,\, 1_{Q.\mathrm{fiber}})} Q$$

is a morphism of $\int_{\mathcal{B}} \mathcal{R}_A$ with base component $(h_1^{-1} \mathbin{⨾} k) \mathbin{⨾} h_2$. Hence P and Q are joined; since they were arbitrary, $\int_{\mathcal{B}} \mathcal{R}_A$ is transitive.

Throughout the calculation z_1 and z_2 are arbitrary C3 coordinates. Their common real value is established in the next step by evaluating the same authored construction at their two semantic inputs and reading its real level. The comparison neither identifies those inputs nor introduces a common source input over 0. $\square$

Theorem 10.8 (Concentricity)**.** *Let A be an A-section. The infinitely many C-residue zero S^6 spheres transported by the A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$, from the projective great-circle base $\mathcal{B}$ to the A-generated normalized value groupoids, form exactly one colimit/component class, and that singleton class carries one real value*

$$c \in \mathbb{R}.$$

Consequently every populated residue-$\mathbb{C}$ zero 6-sphere has centre c, so the infinitely many populated residue-$\mathbb{C}$ zero spheres are concentric.

Proof. The construction stands complete, and the proof runs on it. The base is the projective action groupoid $\mathcal{B}$ (Definition 9.23); conditions C1–C3 built the disk action $D_A = \mathrm{cayleyProjective}(\mathrm{diag}(u_A, 1))$ (Definition 9.7); the representatives and the stabilizer factors carry it over the whole base (Definition 9.14, Lemma 9.15); and the A-section functor gathers the state triples it generates,

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd} \qquad \text{(Proposition 9.20)},$$

with total groupoid

$$\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A \qquad \text{(Proposition 9.22)}.$$

Condition C3 presents the arriving multiplier at N in Weierstrass form (Proposition 10.1, Lemma 9.11), its sphere factors enumerating the residue-$\mathbb{C}$ zeros, and the equation $A = 0$ on the upper half-plane selects the residue states at the one north object. Their preimage under the action — membership closed under composition in the base — is the residue subdiagram $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ (Definition 10.5), with total

$$\int_{\mathcal{B}} \mathcal{R}_A \hookrightarrow \mathcal{T}_A,$$

the inclusion $\int \iota_A$ full and faithful (Lemma 10.6): an isomorphism onto its image, the selected residue states with the restricted transports between them. The proof now spends itself on three steps: transitivity, connectedness, and the readout.

Lemma 10.7 proves, in this exact inverse-image groupoid, that any two objects are joined by one Grothendieck morphism: the north comparison supplies its base component and one element of G_2 its fibre component. Thus $\int_{\mathcal{B}} \mathcal{R}_A$ is connected. This is a statement about the groupoid $\int_{\mathcal{B}} \mathcal{R}_A$: at this stage the established conclusion is connectedness; equality of real coordinates and the common centre arise from the colimit readout below.

By Remark 8.3.5 of [23], a category is nonempty and connected if and only if its π_0 is the singleton set. Condition C4 ensures it is nonempty, since it says there are infinitely many C-residue zeros, and connectedness was just established. Therefore, by Lemma 10.3 (`pi0GrothendieckEquiv`),

$$\pi_0\Big(\int_{\mathcal{B}} \mathcal{R}_A\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}} \,(\pi_0 \circ \mathcal{R}_A)$$

collapses to a singleton κ. The reading is performed by two library theorems, instantiated at the objects just built. A function on the components of $\int_{\mathcal{B}} \mathcal{R}_A$ is a functor from $\int_{\mathcal{B}} \mathcal{R}_A$ to the discrete category on $\mathbb{R}$ (`ConnectedComponents.functorToDiscrete`; Remark 8.3.5 of [23]: π_0 is left adjoint to the discrete inclusion), and on a connected category a functor constant on arrows is constant outright (`constant_of_preserves_morphisms`). The function read is the level the action itself carries: every arrow of $\int_{\mathcal{B}} \mathcal{R}_A$ is an element of the one action with its logarithmic lift, together with one element of G_2; the real part of the lift is constant along the arrow, $\operatorname{Re}\Gamma(0) = \operatorname{Re}\Gamma(1)$, since every branch of the logarithm along a companioned path is $\log|\gamma| + \operatorname{Arg}_{2k}$, so the real part is branch-independent, the one real transport of the degenerate base ([21, Cor. 4.13]); and the certified representatives compute it,

$$\mathrm{transportLevel}_A(n) = \operatorname{Re}\big(A.\,\mathrm{sphereZero}(n)\big).$$

Instantiated at the singleton κ, the two theorems read one real number across the population:

$$\mathrm{transportLevel}_A(n) = \mathrm{transportLevel}_A(0).$$

The proof then sets $c = \mathrm{transportLevel}_A(0)$. Every populated residue-$\mathbb{C}$ zero 6-sphere has real centre c, so the infinitely many residue-$\mathbb{C}$ zero spheres of the A-section are concentric. □

Corollary 10.9 (Translation to the classical framework)**.** *Under the residue-$\mathbb{C}$/classically-nontrivial-zero identification, the already-proved populated residue-$\mathbb{C}$ zero spheres are the classically nontrivial zero spheres and retain the common centre c supplied by Theorem 10.8.*

Proof. This is the translation of the C-residue population through Theorems 8.1 and 8.2. The centre comes directly from Theorem 10.8, while C4 and Lemma 10.2 supply infinitude and disjointness. □

11 Corollaries

Two corollaries close the argument. The first checks that the octonionic zeta is an A-section, verifying its four conditions C1–C4 one by one against the classical facts of Part 1; the Concentricity Theorem then applies to it directly, and its residue-$\mathbb{C}$ zero-spheres share one real centre c. The second corollary is the Riemann Hypothesis. The functional equation of ξ now pins that centre: it sends a zero of real part c to one of real part $1 - c$, and applied to the common centre it forces $c = 1 - c$, so $c = \frac{1}{2}$. Every nontrivial zero lies on the critical line.

Corollary 11.1 ($\zeta_{\mathbb{O}^*}$ is an A-section)**.** *$\zeta_{\mathbb{O}^*}$ is an A-section* (Definition 10.4)*: a section of $\mathcal{R}$ satisfying* (C1)–(C4)*.*

Proof. $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1)** *Meromorphic continuation; simple pole.* Classical fact: ζ is meromorphically continued with a single simple pole at $s = 1$ (Theorem 1.1); the value of $\zeta_{\mathbb{O}^*}$ there is the point at infinity, $\zeta_{\mathbb{O}^*}(1) = \infty = N$ (Definition 4.1).

- **(C2)** *Infinite Euler product.* Classical fact: $\zeta = \prod_p (1 - p^{-s})^{-1} = \exp(\sum_p \ell_p)$ on $\Re s > 1$ with $\ell_p(s) = \sum_{k \geq 1} p^{-ks}/k$ — prime-power logarithmic form with every coefficient $a_{p,k} = 1$, one base, decaying to $\zeta \to 1$ as $\Re s \to +\infty$ (Theorem 1.2).

- **(C3)** *Infinite Weierstrass factorization.* Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]): $(q-1)\,\zeta_{\mathbb{O}^*} = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n)$ (pole factor at $p_0 = 1$; classically, Hadamard factors $(s-1)\zeta(s)$), where $m = 0$ ($\zeta(0) = -\frac{1}{2} \neq 0$), R is the product over the nonzero residue-$\mathbb{R}$ zeros, the classically trivial zeros $-2, -4, \ldots$, honest real zeros of $\zeta_{\mathbb{O}^*}$, and $\{q_n\}$ enumerates its residue-$\mathbb{C}$ zero-spheres.

- **(C4)** *Infinitely many residue-$\mathbb{C}$ zeros.* Classical fact: ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue-$\mathbb{C}$ zeros, so $\{q_n\}$ is infinite.

After the (C1)–(C4) verification, the downstream centre-pinning uses the *functional equation* $\xi(s) = \xi(1-s)$, $\xi(\bar{s}) = \overline{\xi(s)}$ (Theorem 1.1), used in Corollary 11.2. The *residue dictionary*, a closed zero of $\zeta_{\mathbb{O}^*}$ is a real point (residue field $\mathbb{R}$, the classically *trivial* zeros) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$, the classically *nontrivial* zeros), is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes "residue-$\mathbb{C}$" and "nontrivial" the same notion. Thus (C1)–(C4) hold. □

Corollary 11.2 (Riemann Hypothesis)**.** *Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.*

Proof. By Corollary 11.1, $\zeta_{\mathbb{O}^*}$ satisfies (C1)–(C4). By Theorem 10.8, its populated residue-$\mathbb{C}$ value-transport colimit is the real singleton $\{c\}$ and its residue-$\mathbb{C}$ zero spheres are already concentric about c. By Corollary 10.9, these are the classically nontrivial zeros, so every nontrivial zero has real part c. The functional equation $\xi(s) = \xi(1-s)$ (Theorem 1.1) sends a zero of real part c to one of real part $1 - c$; applied to the *common* centre it forces $c = 1 - c$, whence $c = \frac{1}{2}$ (this is Theorem 8.3 read on $\zeta_{\mathbb{O}^*}$). Therefore every nontrivial zero has real part $\frac{1}{2}$, and by (C4) infinitely many lie on $\Re s = \frac{1}{2}$. □

A word on why the argument takes the shape it does. No one can hold an 8-dimensional graph in the mind's eye, or draw the continuum of Riemann spheres the section lives on. The epigraph names the way through: no single viewpoint sees the whole, but when the viewpoints are multiplied and conjugated, a vision appears that none of them held alone. Each slice direction is one such viewpoint; the value-bearing A-section functor gathers all of them over the base, the total object holds them together, and the colimit conjugates them into a single answer. The concentricity we could not picture becomes one component we can compute.

Remark 11.3 (The image span of a fully faithful inclusion)*.* For any fully faithful functor $F : \mathcal{C} \to \mathcal{D}$ there is a canonical roof through the image,

$$\begin{tikzcd}
 & \mathrm{Im}(F) \arrow[dl, "F^{-1}"'] \arrow[dr, hook, "j"] & \\
\mathcal{C} & & \mathcal{D}
\end{tikzcd}$$

the backward leg being an isomorphism of categories onto the image and the forward leg the canonical inclusion. It is a general fact, available here for ι_A and for $\int \iota_A$, and it is the shape a length-two zigzag argument would use. Theorem 10.8 obtains connectedness directly from transitivity at the total, by a single element. The image span records the corresponding length-two alternative.

Remark 11.4 (Two alternative routes). Two other proofs of Theorem 10.8 are available. The first runs over the base. The restriction sits over $\mathcal{B}$ through its two projections,

$$\begin{tikzcd}
\int_{\mathcal{B}} \mathcal{R}_A \arrow[rr, hook, "\int \iota_A"] \arrow[dr, "\pi_{\mathcal{R}}"'] & & \int_{\mathcal{B}} \mathcal{A}_A \arrow[dl, "\pi_{\mathcal{A}}"] \\
 & \mathcal{B} &
\end{tikzcd}$$

and one may connect residue states footpoint by footpoint: zigzags in the connected base, lifted through the fibres. That route travels through $\mathcal{B}$ and reassembles the value data leg by leg. The second is Thomason's homotopy colimit theorem [25]: the classifying space of a Grothendieck construction is homotopy equivalent to the homotopy colimit of the classifying spaces of its fibres, and reading connected components through that equivalence yields the same collapse of $\pi_0\big(\int_{\mathcal{B}} \mathcal{R}_A\big)$. The proof given here stays at the level of categories, where Lemma 10.3 suffices: the base route assembles what the action already supplies whole, the Thomason route is the topological shadow of the same computation, and both remain expository routes outside the Lean development.

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